Beyond All-Equity Lifecycle Portfolios: A Public-Data Multi-Asset Extension

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Abstract

Should households saving for retirement hold only equities? Anarkulova, Cederburg, and O'Doherty (2025) find that internationally diversified stocks outperform conventional balanced and target-date allocations in their baseline model. We recover the domestic-to-international equity ranking in an annual public-data implementation, then widen the investment universe to currency-hedged global bonds, gold, and trend-following futures. Two simple four-sleeve families use fixed weights at exposures from 100% to 200%. Over 1927–2025, the equal-weight rule at 175% exposure reduces simulated retirement ruin from 6.98% to 2.48% and matches benchmark utility with a utility-equivalent saving rate of 6.53% of labor income instead of 10%. At the same 175% exposure, the equity-only benchmark has 14.81% ruin and requires 9.64% saving. The proportional family at 175% requires 5.87% saving and has 2.13% ruin. Post-1970 ablations, historical-panel resampling, and explicit return and financing-cost stresses support the broader diversification result, with greater sensitivity for the equal-weight family. A common planned withdrawal anchored to the equity benchmark favors the proportional family but exposes a limitation of equal weighting. Both families reduce ruin without portfolio-level borrowing, although matching benchmark utility then requires more saving. The evidence supports combining diversification and exposure in lifecycle design, with implementation costs, historical regimes, and the retirement-income objective determining the strength of the gain.

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Introduction

Households saving for retirement face two distinct decisions: which assets to combine and how much exposure to take. Balanced and target-date portfolios often change both at once. Adding bonds reduces the equity share and usually reduces total volatility, so the effects of composition and exposure arrive together. Lifecycle models place these choices in a household setting with labor income, longevity risk, retirement consumption, and bequests (e.g., Campbell and Viceira, 2002; Cocco, Gomes, and Maenhout, 2005). Anarkulova, Cederburg, and O'Doherty (2025), hereafter ACO, find that a fixed allocation of approximately 33% domestic and 67% international equities outperforms conventional balanced and target-date strategies in their baseline setting. Their opportunity set contains domestic and international stocks, local sovereign bonds, and local bills. Their leverage sensitivities also show that financing costs influence the optimal allocation.

How far does the equity result extend? ACO diversify stocks internationally but their baseline bond allocation holds local sovereign debt. We broaden the bond sleeve across issuers and add gold and trend-following futures. These assets bring different exposures to inflation, currency movements and market trends. Whether those differences improve retirement outcomes depends on their joint return distribution and on the cost of holding them.

The economic rationale is to combine assets before deciding how much to borrow. In Markowitz (1952), imperfectly correlated returns can reduce portfolio variance for a given expected return. Asness (1996) applies this logic to a levered balanced portfolio as an alternative to an all-equity allocation. Beyond variance, Moskowitz, Ooi, and Pedersen (2012) find that diversified trend-following futures performed particularly well during extreme market movements. These findings motivate testing both annual dispersion and retirement shortfalls; they do not establish that our portfolios have smaller drawdowns or dominate the entire return distribution. Here, an improvement means a lower simulated probability of exhausting retirement assets and a lower saving rate needed to attain the benchmark's expected utility. Annual volatility is reported separately. We also scale ACO's equity allocation through the same exposure ladder, financing each borrowed dollar at the resident bill plus the same spread. This separates the benefit of broader diversification from the effect of borrowing more against equities.

Our annual replication uses the Jordà–Schularick–Taylor Macrohistory Database and supplementary public sources.[1] The final panel contains 1,561 country-years across 16 developed markets over 1927–2025. It recovers the qualitative ranking of the fixed equity and balanced strategies: simulated retirement ruin is 6.98% for the 33/67 benchmark, compared with 15.52% for domestic equities. The 33/67 mix is also near the utility maximum on our public-panel domestic–international equity grid: the best mix, 25/75, requires 9.96% saving to match the utility of 33/67 at 10%. The multi-asset comparison therefore starts from a nearly optimal unlevered equity mix within that grid.

The central experiment keeps the benchmark's equity mix fixed and adds currency-hedged

[1]The research repository is available at `https://github.com/GF-boop/beyond-stocks`. Numerical outputs, replication scripts, and data provenance are detailed in Appendix G.

global bonds, gold, and managed futures in two simple families. The proportional family starts from a conventional 60/40 stock–bond portfolio and borrows 50% of investor capital to add 25% gold and 25% managed futures, giving 60/40/25/25 at 150% gross exposure. We scale all four positions proportionally to obtain the other exposure levels. The equal-weight family assigns one quarter of gross exposure to each sleeve. Equal weighting avoids estimating allocation weights from return moments; DeMiguel, Garlappi, and Uppal (2009) find that none of the 14 portfolio models they examine consistently outperforms $1/N$ out of sample across seven datasets. This motivates a transparent benchmark, without implying that equal weighting is optimal for our four sleeves. Both families use fixed exposures of 100, 125, 150, 175, and 200%.

The full-panel equal-weight allocation at 175% exposure reduces simulated ruin from 6.98% to 2.48% and equivalent saving from 10% to 6.53%. The raw annual volatilities are close, at 25.54% and 25.28%, respectively, but extreme currency conversions strongly influence these estimates: Italy 1942 alone accounts for 48.35% of the equity benchmark's pooled sum of squared return deviations. We therefore examine exclusions of the largest source events, Italy 1942 and France 1946. Removing Italy 1942 gives 17.64% volatility against 18.17%, 2.42% ruin against 7.19%, and 6.41% equivalent saving against 10%. Removing France 1946 as well gives the same direction on all three measures. The ranking also survives a broader historical-access and monetary-stress screen: equal-weight 175% has 15.17% volatility, 0.57% ruin, and 5.93% equivalent saving, against 17.02%, 5.04%, and 10% for ACO. This screen excludes documented disruptions, extreme inflation, and the influential Italy observation; it is a retrospective sensitivity, not proof that every retained return was historically investable.

The primary sample covers 1927–2025, retaining the range of monetary and market regimes in the historical record. Post-1950 and post-1970 windows examine the contribution of early history and preserve the favorable ruin and utility ranking of the two 200% families. Thus the lower ruin and saving requirements are not confined to the early historical sample. The central question is whether these gains from broader diversification persist across exposure levels, portfolio compositions, and implementation assumptions.

Section 5 distinguishes these household-level gains from adoption across the retirement system: large reallocations can change prices and returns in equities and gold, while common trend-following signals raise additional questions about trading capacity.

1 ACO and the Multi-Asset Extension

1.1 The benchmark result

ACO evaluate retirement strategies in a lifecycle model with stochastic labor income, Social Security benefits, mortality risk, and a bequest motive. A couple works for 40 years and then draws down its retirement account. Their monthly return panel covers 39 developed countries over 1890–2023, with coverage varying by country. Multi-year bootstrap blocks can originate in different countries, while household earnings, demographics, and preferences remain anchored to a U.S. baseline.

Their baseline fixed allocation holds approximately 33% domestic and 67% international equities without borrowing. ACO also relax the leverage constraint: the optimal exposure and bond allocation depend on the borrowing spread. Our extension examines additional asset classes under explicit financing assumptions, whose sensitivity is reported in Section 4.

ACO select portfolio weights by maximizing average simulated utility. On our public panel, a grid search restricted to the domestic–international equity mix places the optimum at 25% domestic and 75% international, with 33/67 within 0.04 percentage points of equivalent saving. We therefore retain 33/67 as a common equity sleeve and reference strategy. The multi-asset families use fixed heuristic weights rather than an analogous optimization. The resulting comparison asks whether simple diversified rules improve on the equity allocation selected in this framework.

Expected utility already trades off favorable outcomes against adverse ones under the household's preferences. Comparing portfolios at similar annual volatility adds a complementary diagnostic: it helps distinguish changes in composition from changes in overall exposure. We use the fixed ladders of Section 4 for that purpose. The ladders report annual volatility alongside the lifecycle outcomes at each exposure level. Appendix E separately examines how the comparison changes with risk aversion.

The question is what investors today can learn from long-run market history, not whether they could have purchased the same funds in 1927. Managed-futures ETFs and funds combining traditional assets with managed-futures exposure now provide practical ways to implement these allocations.[2] Historical availability of the fund wrapper is therefore not the relevant criterion: a globally diversified equity portfolio evaluated over ACO's history likewise differs from what households could readily access at the time. Both exercises use historical returns to inform present-day allocation decisions.

1.2 The central question

Having recovered ACO's main portfolio ranking on public data, we ask whether diversification across asset classes can further reduce retirement ruin and the saving required to attain benchmark utility, including when the equity benchmark uses the same leverage.

Section 4 compares the fixed exposure ladders, removes individual sleeves to identify their contributions, and tests sensitivity to historical windows, household preferences, and implementation costs. A common retirement-income target also checks whether lower ruin under each portfolio's own withdrawal rule translates into a greater ability to fund the same planned spending.

[2]Examples include the iMGP DBi Managed Futures Strategy ETF (DBMF) and the Return Stacked U.S. Stocks & Managed Futures ETF (RSST). These illustrate available implementation instruments; access depends on jurisdiction and brokerage arrangements, and their exposures and costs need not match the simulated portfolios exactly.

2 Data

2.1 Country-year panel and resident numeraire

All empirical inputs to our lifecycle simulations are publicly accessible. Whereas ACO evaluate their model on proprietary monthly return series, we reconstruct the long-run developed-market record from open sources, ensuring that the replication and every extension can be reproduced from public data alone. The only proprietary series in the paper are the SG and BarclayHedge managed-futures indexes, which appear solely in the external validation of Appendix B.2 and enter no lifecycle simulation.

The panel begins with release 6 of the Jordà–Schularick–Taylor Macrohistory Database (Jordà et al., 2023; Jordà et al., 2019), which supplies nominal total returns on equities and long-term government bonds, short rates, inflation, exchange rates, and real GDP for 16 developed markets through 2020. We extend each series through 2025 using dividend-adjusted local indices, locally listed ETFs, and the Global Macro Database (Müller et al., 2025). Over the 2006–2020 overlap, the reconstructed equity series average approximately 1.2 percentage points below the Macrohistory series, primarily reflecting ETF management fees and large-cap benchmark tilts. A source indicator flags the appended years, allowing robustness checks to isolate the original Macrohistory window. The resulting baseline dataset is an unbalanced panel of 1,561 country-years.

A historical market-access and monetary-stress screen is retained as a labeled sensitivity. It excludes calendar years overlapping the prolonged closures or restrictions reported for our countries in ACO's Table A.III and years with CPI inflation of at least 50% or deflation of at least 20% in absolute value. These criteria flag potential obstacles to market access and currency conversion; they do not establish non-investability for every flagged observation. Annual realized inflation makes the screen retrospective. The source-screened comparison additionally removes Italy 1942, selected through the variance audit, from resident rows and global equity and bond baskets. The replication package preserves the raw rows. Every calendar year from 1927 through 2025 remains represented.

Let $S_{c,f,t}$ denote units of resident currency c per unit of foreign currency f. If $R^N_{f,t}$ is a nominal foreign return and $\pi_{c,t}$ is resident inflation, the resident-real return is

$$1 + R^{(c)}_{f,t} = (1 + R^N_{f,t})\frac{S_{c,f,t}}{S_{c,f,t-1}}\frac{1}{1+\pi_{c,t}}. \tag{1}$$

We apply this spot conversion to unhedged assets before aggregating returns. Domestic equities, international equities, gold, and unhedged controls therefore all measure changes in purchasing power within the resident household's currency.

International equity excludes the country of residence and uses lagged market capitalization weights constructed from the Big Bang Database's historical market-capitalization-to-GDP ratios when coverage is sufficient (Kuvshinov and Zimmermann, 2022), with real-GDP weights (Maddison Project via the Macrohistory Database, extended by Müller et al. 2025) as a fallback. The

33/67 benchmark combines the resident's domestic market with this foreign basket. The constant-real-exchange-rate counterfactual instead aggregates each foreign market's local real return with identical weights. This accounting counterfactual excludes currency carry and hedging costs; it isolates the contribution of real exchange-rate movements.

The world bond block equally weights every sovereign in the panel with a usable bond, bill, and exchange rate in a given year. In 1,344 of 1,561 country-years it holds all 16 issuers; the floor anywhere is 13. Years with fewer than eight available issuers are dropped, which costs no observation after 1927. Like an actual global bond index, the block includes the resident's own sovereign. The basket is therefore common to all residents in a year, and each foreign bond uses the initial-notional forward of equation (4). Financing stays local throughout. We subtract ten basis points annually from covered notionals in the baseline. The unhedged foreign bond and bill basket is retained only as a labeled diagnostic control. The resulting basket spans the developed sovereign markets in the panel.

2.2 Gold and managed futures

Gold uses the annual U.S.-dollar price compiled by MeasuringWorth (Officer and Williamson, 2024) through 1967 and the LBMA afternoon fixing (London Bullion Market Association, 2025) from 1968, the first year a free market existed. Both are dollars per troy ounce, spliced without recalibration. We subtract 40 basis points per year for custody, convert the dollar return into each resident currency, and deflate by resident inflation. Before 1968, the dollar price was largely administered, and private U.S. ownership and convertibility were restricted. Under the gold standard and Bretton Woods, exchange-rate parities also constrained gold prices in other panel currencies. The early low-variance segment therefore reflects a monetary regime distinct from the subsequent free market (Erb and Harvey, 2013). The post-1970 window examines the portfolio comparison after this transition.

No free investable managed-futures index spans the sample. We reconstruct a monthly time-series-momentum proxy instead, following a well-established literature. Moskowitz, Ooi, and Pedersen (2012) document that the sign of an asset's own past twelve-month return predicts its next-month return across roughly 58 futures markets, and that scaling each position by inverse volatility and averaging produces a diversified premium. Hurst, Ooi, and Pedersen (2017) extend the same rule back to 1880 and show that a blend of one-, three-, and twelve-month look-backs is stable across a century of data and across asset classes. Nick Baltas and Kosowski (2020) turn to the construction choices themselves. Signal horizon, rebalancing frequency, and the resulting turnover jointly determine net performance and capacity. Hence our one-, six-, and twelve-month blend, and hence the explicit turnover cost below. Lempérière et al. (2014) find a trend premium in equities, bonds, currencies, and commodities as far back as 1800. That history motivates evaluating the rule over the full 1927–2025 window. Trend following also lowers sequence risk in retirement decumulation (Clare et al., 2017; Clare et al., 2019) and improves risk-parity implementations through correlation-aware weighting (Nikitas Baltas, 2015). These studies already establish a retirement-income motivation for trend following. Our contribution is to evaluate this mechanism alongside global bonds and gold in an annual adaptation of ACO's household framework, reporting utility-equivalent saving as well as capital depletion under explicitly matched portfolio financing.

The managed-futures signal uses a separate, frozen monthly price snapshot. Completed calendar-year returns are compounded and joined to the annual JST/Macrohistory household panel. Appendix B.1 documents that snapshot, its sources, and the monthly-to-annual transformation in detail.

In our implementation, signals average the signs of prior one-, six-, and twelve-month returns. Volatility comes from the preceding 36 months. The aggregate strategy targets 10% volatility under a three-times scalar cap. The universe expands across four sectors as data become available: 18 national price indexes and 18 synthetic ten-year sovereign positions, 19 commodities plus gold and silver, and six dollar foreign-exchange forwards. The monthly equity and yield inputs come principally from the OECD Main Economic Indicators via FRED (Organisation for Economic Co-operation and Development, 2026; Federal Reserve Bank of St. Louis, 2026). The early U.S., U.K., German, French, commodity, and rate segments come from the monthly NBER Macrohistory files (National Bureau of Economic Research, 2016), commodities are extended with the World Bank "Pink Sheet" (World Bank, 2025), and currencies use month-end Federal Reserve H.10 quotations (Board of Governors of the Federal Reserve System, 2026). Where two sources meet, the return that straddles the junction is left missing. This prevents source-level differences from entering the trading signal.

We use the reconstructed managed-futures series to extend the public-data allocation experiment to trend following. Its returns include U.S.-dollar collateral income and are converted into the household's currency using the same initial-notional hedge convention as global bonds. Appendix B.2 reports the external comparisons; Appendix B.1 documents the construction details and limitations.

We subtract a 0.85% annual management fee and transaction costs based on measured turnover and a three-basis-point spread. Its monthly correlation with several official CTA indexes is about 0.55–0.59 over 2000–2025.

The additional return-deduction experiment in Appendix L measures how much deterioration in the proxy's returns the portfolio results can absorb.

2.3 Realized portfolio properties

Table 1 reports pooled one-year return moments from the same panel and hedge convention as the lifecycle results. The table reports mean returns, volatility, and the lower annual tail. Global bonds have slightly higher volatility than local bonds in the 90/60 comparison under this convention.

Table 1: Realized resident-real portfolio properties

Portfolio	Mean return	Volatility	1% annual return
Domestic equity	7.60%	23.34%	-45.64%
ACO 33/67	8.12%	25.28%	-40.36%
Stocks/I 50/50	7.99%	22.59%	-40.22%
Balanced domestic	5.38%	15.82%	-34.19%
Balanced/I	5.61%	15.04%	-27.63%
90/60 local	8.25%	23.95%	-36.55%
90/60 global bonds	8.19%	24.13%	-36.19%
ACO, constant real FX	7.44%	15.75%	-37.40%
60/40 ACO/covered	5.64%	16.51%	-25.67%
60/40 ACO + 33.33 MF	7.91%	17.95%	-20.08%
60/40 ACO + 33.33 gold	6.79%	23.86%	-23.33%
54/36/20/20 ACO	7.18%	19.94%	-19.08%
90/60/25/25 ACO	10.75%	30.34%	-29.39%

Note: Arithmetic means, standard deviations and linearly interpolated 1% quantiles of pooled annual returns across 1,561 country-years. Tabulated from the baseline simulation outputs. Bonds and managed futures use the fixed-notional hedge; managed futures include baseline fees and costs.

3 Lifecycle Model and Portfolio Design

3.1 Household, income, and retirement

The household is a couple aged 25. Each spouse receives an independent stochastic earnings history through age 64 based on model 6 of Guvenen et al. (2021). Log income contains a quadratic age profile, a persistent component with autoregressive coefficient 0.959, transitory shocks, and age- and income-dependent nonemployment. The household saves a constant fraction of each spouse's income when that individual's annual income is at least $15,000. Contributions arrive at the start of the year, in time to earn that year's portfolio return.

At age 65 the couple fixes a real annual withdrawal equal to 4% of retirement wealth, following Bengen (1994) and ACO. A fixed real withdrawal makes the retirement outcome sensitive to the order in which returns arrive, not only to their average: a sequence of early losses can exhaust the account even when long-run returns are adequate, and international evidence shows the safe rate varies widely across historical return paths (Estrada, 2018). Appendix F varies the withdrawal rate from 3% to 5% and the contribution rate from 5% to 15% and finds the diversified ranking unchanged. Social Security follows from each simulated earnings history, computed with the 2022 $147,000 taxable-earnings ceiling, 2022 bend points, spouse and survivor benefits, and the early-claiming adjustments ACO describe. Supplemental Security Income supplies a consumption floor. Ruin occurs when financial wealth cannot fund the fixed withdrawal before the last spouse dies. Public pension income continues after financial wealth is exhausted.

Mortality follows sex-specific Gompertz–Makeham laws calibrated to the joint retirement and longevity moments reported by ACO from Social Security Administration tables (Social Security

Administration, 2020). Return paths, income histories, and death draws are paired across strategies.

Following ACO, the model combines U.S. earnings, public pensions, and preferences with pooled developed-market financial history. Each simulated lifetime draws successive blocks from this history, exposing the same household to a wider range of economic environments than a single national record provides. Countries share shocks and global asset baskets, so the 1,561 country-years form a dependent panel. Every strategy faces the same return, income, and mortality draws within an experiment. Section 3.4 describes the resampling procedure.

3.2 Utility and equivalent saving

For annual real consumption C_t, household size H_t, bequest B, and last year alive T, utility is

$$U(C,B) = \sum_{t=65}^{T} \frac{(C_t/\sqrt{H_t})^{1-\gamma}}{1-\gamma} + \theta \frac{(B+k)^{1-\gamma}}{1-\gamma}. \tag{2}$$

Risk aversion is $\gamma = 3.84$, the bequest motive has intensity $\theta = 2{,}360$, and the shifter is $k = \$490{,}000$. These are the estimates of De Nardi, French, and Jones (2010), fitted to retiree data in a lifecycle model with medical expenses, and ACO adopt them unchanged. Our utility aggregator therefore uses their stated preference parameters, while the wealth and consumption dynamics remain an annual approximation of their monthly engine. The bequest term is additive and warm-glow: utility depends on the size of the estate left, $B + k$, not on the heirs' own circumstances, which is the specification that explains why retirees in the data run down wealth slowly. ACO evaluate monthly consumption and therefore multiply the annual bequest-intensity estimate by 12^{γ}. If within-year consumption is uniform, dividing their entire monthly utility by 12^{γ} gives equation (2) with $\theta =$ 2,360. Under uniform within-year consumption, the annual aggregation thus preserves the relative scale of consumption and bequest utility.

For strategy j, the equivalent saving rate s_j^* solves

$$E[U_j \mid s_j^*] = E[U_{33/67} \mid s = 0.10]. \tag{3}$$

A rate below 10% means the strategy reaches benchmark utility on less saving. Every other reported outcome, wealth, consumption, ruin, and bequest, uses the common 10% rate and not s_j^*. The metric evaluates composition and exposure jointly. A lower-exposure portfolio can require more saving despite reducing the probability of ruin. The fixed-exposure ladders in Section 4 and Appendix C therefore report utility alongside annual volatility. The joint comparison identifies what additional exposure contributes to retirement outcomes.

Higher risk aversion penalizes low household consumption rather than annual return variance itself. Social Security and the consumption floor moderate the transmission of portfolio losses to consumption, while large positive returns can raise measured volatility without creating low-consumption states. Equivalent saving also compares each rule with a benchmark whose utility is recomputed at the same γ. With annual bequest intensity held fixed, raising γ from 2 to 10

Table 2: Fixed portfolio rules

Portfolio	Equity	Bonds	Gold	MF	Sleeve sum	Financing
ACO 33/67	100%[a]	0%	0%	0%	100%	0%
90/60 local	90[a]	60[b]	0	0	150	50
90/60 global bonds, covered	90[a]	60[c]	0	0	150	50
60/40 ACO/covered	60[a]	40[c]	0	0	100	0
60/40 ACO + 33.33 MF	60[a]	40[c]	0	33.33	133.33	33.33
60/40 ACO + 33.33 gold	60[a]	40[c]	33.33	0	133.33	33.33
54/36/20/20 ACO	54[a]	36[c]	20	20	130	30
90/60/25/25 ACO	90[a]	60[c]	25	25	200	100

Note: Percentages of investor capital. [a]33% domestic and 67% international equities. [b]Residence-country government bond and bill. [c]Currency-hedged equal-weight global sovereign bonds with resident-currency financing. The equity sleeve remains 33/67. "Sleeve sum" is the sum of index-level allocations, and financing is the corresponding external cash financing in the return equation.

increases required saving from 3.17% to 4.84% for the equal-weight 200% rule, but from 9.23% to 17.60% for ACO at the same exposure. The response therefore depends on the consumption distribution, not on leverage alone (Appendix E).

3.3 Fixed, non-estimated portfolio rules

Table 2 defines the main extensions. The 90/60 local strategy holds 90% in the ACO 33/67 equity portfolio and 60% in the residence-country sovereign bond, financed with a 50% short local-bill position plus the financing spread ϕ on the borrowed portion. The intermediate strategy changes only the bond sleeve to the covered global aggregate, with financing still in the resident bill. Every other extension also uses the same 33/67 resident equity return. The designs therefore vary fixed income, alternative sleeves, and total exposure without changing equity geography.

The weights are design inputs fixed prior to evaluation.

The manifest also contains two fixed leverage ladders, each stepping a four-sleeve family through 100, 125, 150, 175, and 200% gross exposure: one proportional to 60/40/25/25 (equity/covered bonds/gold/managed futures), the other equal-weight across the four sleeves. The levels are fixed before evaluation and target no realized volatility. These ladders carry the central comparison in Section 4.

Let A_t^f be the foreign asset's gross nominal return, B_t^f and B_t^c the foreign and resident gross nominal bill returns, and $q_t = S_{c,f,t}/S_{c,f,t-1}$ the spot currency movement. The baseline covers the initial foreign principal with a one-period forward. Under the assumed covered-interest-parity forward ratio, its resident-real return is

$$1 + R_t^{c,H} = \frac{B_t^c/B_t^f + (A_t^f - 1)q_t}{1 + \pi_{c,t}}. \tag{4}$$

The asset gain or loss beyond the initial principal retains currency exposure. Annual realized bill returns proxy the interest rates used to price the forward. The former full-value hedge is retained only as a sensitivity. Let every $R_{i,t}^{(c)}$ then be in the common resident-real numeraire. A portfolio whose index-level sleeve sum is G_j has return

$$R_{j,t}^{(c)} = \sum_i w_i R_{i,t}^{(c)} + (1 - G_j) R_{f,t}^{(c)} - \max(G_j - 1, 0)\phi - h_j \kappa, \tag{5}$$

where ϕ is the financing spread, h_j is covered notional, and κ is the hedge friction. The managed-futures NAV includes its recorded U.S. collateral once; the forward payoff is added under equation (4). The portfolio financing position in equation (5) is separate. For ACO at exposure G, the same equation becomes $R_{\text{ACO},G} = GR_{33/67} + (1 - G)R_f - (G - 1)\phi$ for $G \geq 1$. Its equity sleeve remains unhedged. We evaluate $G = 1, 1.25, 1.5, 1.75, 2$; the utility reference remains unlevered ACO at a 10% saving rate.

For a levered portfolio ($G_j > 1$), the coefficient on the resident bill in equation (5) is negative. Holding all asset returns fixed, a higher ex-post real bill therefore lowers the portfolio return, while an inflation surprise that lowers the ex-post real bill raises it. This effect follows directly from the accounting identity of annual rebalancing, abstracting from intra-year borrowing constraints.

The baseline financing and hedge charges are 30 and 10 basis points, respectively. Both enter as annual real-return deductions. Cost sensitivities vary these deductions while retaining the observed real bill return.

Gross exposure sums allocations to return sleeves; managed futures also uses variable derivative positions within its NAV. The annual model captures portfolio financing and hedge costs through equation (5).

Return-stacked ETFs now give individual investors access to combined exposures through ordinary fund shares. For example, the Return Stacked Global Stocks & Bonds ETF targets approximately one dollar of global equity exposure and one dollar of U.S. Treasury exposure per dollar invested (Tidal Trust II, 2026). Funds combining bonds or equities with managed futures provide further building blocks (Tidal Trust II, 2024). These structures bring portfolios of the kind studied here within reach without requiring the household to manage derivative positions or borrowing directly. When shares are purchased outright, without borrowing or pledging them as collateral, the investor faces no personal margin calls arising from the fund's leverage. The fund manages collateral and financing internally; the associated costs and any deleveraging affect its net asset value. Actual funds differ in their asset coverage and trading rules from the synthetic sleeves used here.

Retail investors can access financing substantially cheaper than standard broker margin loans. Return-stacked funds obtain embedded financing through futures; industry evidence for S&P 500 and Treasury futures places this funding near short-term institutional rates (Return Stacked Portfolio Solutions, 2024). An eligible investor can also borrow by selling a box spread on European-style index options, receiving cash against a fixed terminal payment (CME Group, 2024). A practitioner article published by Cboe reports SPX box financing around 30–50 bp above maturity-matched

Treasury yields for three- and five-year maturities, using February 2025 market estimates (Wang, 2025). This evidence supports the order of magnitude of our 30 bp baseline as an efficient retail-financing assumption, rather than an institutional-only privilege. It does not establish a current executable 30 bp quote for every maturity, currency, or account; commissions, execution, fund fees, and collateral requirements must be accounted for separately. Box maturities also differ from our annually reset bill financing.

Relative to these efficient funding routes, a persistent 100 bp spread is already an expensive financing scenario for a cost-conscious investor with the requisite access; 200–300 bp represents a severe stress. Standard margin loans remain a more costly alternative: the IBKR schedule consulted for this revision quotes benchmark plus 150 bp for its initial Pro tier and plus 250 bp for Lite (Interactive Brokers, 2026). These broker markups use a different reference rate from our resident bill. The financing sensitivities therefore measure robustness to paying more for exposure, rather than assuming that retail investors must incur those higher spreads.

A complementary diagnostic examines direct borrowing in a personal margin account. With annual rebalancing and a 25% maintenance requirement, none of the four tested allocations breaches the threshold at a year-end across 10,000 simulated 86-year return paths, or 860,000 account-years per allocation. The minimum equity-to-assets ratio is 48.91% for the equal-weight 175% portfolio and 38.59% and 40.39% for the proportional and equal-weight 200% portfolios. Appendix H reports the calculation, including the 150% stock–bond comparator. It also provides a monthly USD proxy control over 1970–2025, with positions held between annual rebalances. Monthly monitoring identifies breaches that a December-only test misses for levered ACO. Neither diversified family crosses the 25% threshold in that control, including at 200% exposure. The same test records 1,081 threshold breaches for ACO at 175% and 17,934 at 200%, corresponding to 0.13% and 2.09% of simulated account-years. These are threshold diagnostics under annual exposure resets, without modeling forced liquidation after a breach.

3.4 Bootstrap and inference

The main fixed-portfolio experiment draws 10,000 paths using a stationary block bootstrap with mean block length ten years, following ACO. Ten-year blocks preserve two things an i.i.d. resampling would distort: the horizon-dependent structure of equity returns, including the long-run mean reversion documented by Poterba and Summers (1988), and the persistence of real interest rates. The block advances through consecutive years of a country's history, preserving the joint asset-return vector. A completed block starts again at a uniformly sampled panel row. At a gap or series end, the unfinished block continues at the first available year of another country. The implementation selects that country through a uniformly sampled row, so countries are weighted by their available sample lengths. This implements ACO's continuation principle at annual frequency.

Our resident-numeraire conversion operates inside each block. Every asset on a country-year row already sits in that country's currency and price index (Section 2). A block drawn from, say, Japan in the 1980s is therefore a fully consistent yen-real decade: domestic and foreign equity, the hedged bond and managed-futures blocks, and inflation all measured in one household's purchasing power. The numeraire changes only when the next block comes from another country.

We report normal 95% intervals for the paired difference in ruin relative to the 33/67 benchmark. These intervals measure Monte Carlo uncertainty conditional on the observed panel. Appendix D separately reports the range of outcomes across historical windows and source-country omissions. We also resample complete calendar-year cross-sections in outer stationary blocks averaging 5, 10, or 20 years. Each of 100 outer histories per setting is evaluated on 1,000 paired inner lifecycle paths. This preserves the within-year cross-country and cross-asset dependence, including common global sleeves. The resulting percentile ranges describe variation across resampled histories together with residual inner Monte Carlo noise; they are not calibrated confidence intervals. Appendix D details the construction and its limits.

4 Results

4.1 Replication and the central comparison

The annual panel contains 1,561 country-years. We use the stationary bootstrap with the continuation rule described by Anarkulova, Cederburg, and O'Doherty (2025), retain the assembled historical observations, including flagged observations, and value the currency hedge as a one-period, fixed-notional forward. The baseline applies no return-dependent screen.

Table 3 gives the primary results. The public-data replication preserves the economic direction of the international-equity comparison: ACO 33/67 has lower simulated ruin than domestic equity. The equal-weight four-sleeve rule at 175% exposure reduces simulated ruin from 6.98% to 2.48% and utility-equivalent saving from 10% to 6.53%. The paired 95% Monte Carlo interval for the ruin difference is $[-4.97, -4.03]$ percentage points. Annual volatility is 25.54%, compared with 25.28% for ACO; the source-influence diagnostics below examine the concentration of these full-panel estimates.

The complete comparison in Table 3 reports both the raw and source-screened panels for every allocation and exposure. Large converted returns make raw annual variance an unreliable basis for identifying a common risk budget. We therefore assess annual dispersion under both panels, with the full return moments and a separate winsorization check in Appendix I. The two conventions retain the lifecycle advantage of diversification, without requiring equality of raw volatility.

4.2 Diversification and equity leverage

Increasing ACO's equity exposure initially lowers its saving requirement: the 125% and 150% rules require 8.56% and 8.30%, respectively. Beyond 150%, the saving requirement rises to 9.64% at 175% and 15.66% at 200%. Ruin increases throughout the ladder, from 6.98% at 100% to 20.28% at 200%.

At the same 175% gross exposure, the proportional and equal-weight portfolios have ruin of 2.13% and 2.48%, against 14.81% for levered ACO, and require 5.87% and 6.53% saving, against

Table 3: Complete raw and source-screened portfolio comparison

Portfolio	Raw vol.	Raw ruin	Raw saving	Screen vol.	Screen ruin	Screen saving
Domestic equity	23.34%	15.52%	18.60%	22.70%	10.87%	14.47%
ACO 33/67	25.28%	6.98%	10.00%	17.02%	5.04%	10.00%
Stocks/I	22.59%	7.78%	10.56%	17.61%	5.30%	10.13%
Domestic balanced	15.82%	13.65%	19.95%	15.02%	7.17%	14.31%
Balanced/I	15.04%	9.10%	14.81%	12.06%	4.15%	12.45%
90/60 local	23.95%	7.27%	9.72%	16.97%	4.50%	9.01%
90/60 global bonds	24.13%	6.91%	9.49%	16.47%	4.26%	8.88%
ACO, constant real FX	15.75%	4.32%	9.24%	15.66%	3.49%	9.52%
ACO 33/67 125%	31.47%	8.13%	8.56%	21.15%	6.92%	9.23%
ACO 33/67 150%	37.71%	10.69%	8.30%	25.33%	10.39%	9.64%
ACO 33/67 175%	43.99%	14.81%	9.64%	29.53%	14.33%	12.35%
ACO 33/67 200%	50.28%	20.28%	15.66%	33.74%	21.09%	25.14%
60/40 ACO/covered bonds	16.51%	8.45%	14.20%	11.34%	3.84%	12.35%
60/40 ACO + 33.33 MF	17.95%	2.75%	7.61%	12.69%	0.55%	6.82%
60/40 ACO + 33.33 gold	23.86%	4.55%	10.87%	13.18%	1.80%	10.04%
54/36/20/20 ACO	19.94%	3.10%	9.03%	11.92%	0.68%	8.16%
Proportional 100%	15.66%	5.04%	12.35%	9.43%	1.21%	10.76%
Proportional 125%	19.23%	3.32%	9.46%	11.44%	0.72%	8.49%
Proportional 150%	22.88%	2.53%	7.39%	13.53%	0.55%	6.81%
Proportional 175%	26.58%	2.13%	5.87%	15.66%	0.47%	5.51%
Proportional 200%	30.31%	1.80%	4.72%	17.81%	0.51%	4.52%
Equal-weight 100%	15.09%	5.90%	13.26%	9.19%	1.67%	11.33%
Equal-weight 125%	18.50%	4.00%	10.25%	11.12%	0.96%	9.02%
Equal-weight 150%	21.99%	2.97%	8.11%	13.12%	0.69%	7.28%
Equal-weight 175%	25.54%	2.48%	6.53%	15.17%	0.57%	5.93%
Equal-weight 200%	29.12%	2.27%	5.31%	17.25%	0.61%	4.89%
90/60/25/25 ACO	30.34%	2.08%	4.90%	18.27%	0.73%	4.75%

10,000 paths per panel, seed 20260827, 1927–2025. Screened sources exclude documented disruptions, inflation outside $(-20\%, 50\%)$, and Italy 1942 identified by the audit. The same exclusions apply to resident rows and reconstructed equity and bond baskets. The screen is retrospective and does not establish a historically implementable trading rule. MF inputs are unchanged. Saving matches each panel's own unlevered ACO at 10%; cross-panel saving levels therefore have different utility targets. Common seeds pair strategies within each panel, not identical paths across different panel supports.

9.64%. At 200%, their saving requirements fall further to 4.72% and 5.31%, while ACO's rises. Thus the gains cannot be reproduced simply by applying the same leverage to the equity benchmark. Equal gross exposure equates portfolio borrowing; annual volatility remains an outcome of composition. Appendix J reports the complete ACO ladder and the matched-exposure construction, cost, and household-policy sensitivities.

4.3 Which sleeves account for the result?

We examine whether the advantage over ACO extends to simpler compositions by removing global bonds, gold, and managed futures one at a time from each family at gross exposures of 100, 125, 150, 175, and 200%. The removed notional is reallocated pro rata among the three remaining sleeves, leaving gross exposure, borrowing, hedge convention, and per-unit cost assumptions unchanged. Total sleeve and hedge charges adjust with the resulting weights. The comparison measures the effect of changing composition at fixed exposure.

Figure 1 uses a demanding, historically informed benchmark. ACO's 33/67 equity allocation was selected by maximizing simulated lifecycle utility in their historical framework; in our public-panel equity grid, it is within 0.04 percentage points of equivalent saving of the best domestic–international mix. The diversified sleeve weights and their pro-rata ablations are fixed rules, without a corresponding optimization of historical utility. Their sizeable gains at 175% exposure therefore demonstrate that expanding the asset opportunity set can improve on an already well-chosen equity allocation using simple portfolio rules. This comparison does not constitute an out-of-sample test: all strategies are evaluated on historical data, and the ACO 33/67 mix is held fixed across the leverage ladder rather than reoptimized at each exposure.

At 175% exposure, every three-sleeve allocation in Figure 1 has lower simulated ruin than ACO's 6.98%. Five of the six also match benchmark utility with less than 10% saving. Across the proportional ablations, ruin ranges from 1.92% to 3.71% and equivalent saving from 4.15% to 7.79%. In the equal-weight family, the versions without global bonds or gold also improve both outcomes. The version without managed futures reduces ruin to 5.57%, while requiring 10.14% saving to match benchmark utility. The lifecycle gains therefore extend across several compositions and remain visible when any one of the three additional sleeves is removed.

The comparison with the complete portfolios clarifies the roles of the individual sleeves. Managed futures contribute most to capital preservation: including the proxy lowers ruin by 1.58 and 3.09 percentage points relative to the respective reallocations without it. Global bonds reduce annual volatility relative to both reallocations without them. The equal-weight version without gold improves all three reported metrics relative to its complete portfolio, illustrating that the simple four-sleeve rules leave room for alternative compositions. These comparisons measure each sleeve together with the specified reallocation of its weight; their effects are not additive contributions to the total gain over ACO. All six ablations also have lower ruin and annual volatility than ACO at 175% exposure. Five require less saving than its 9.64%; the equal-weight reallocation without managed futures requires 10.14%.

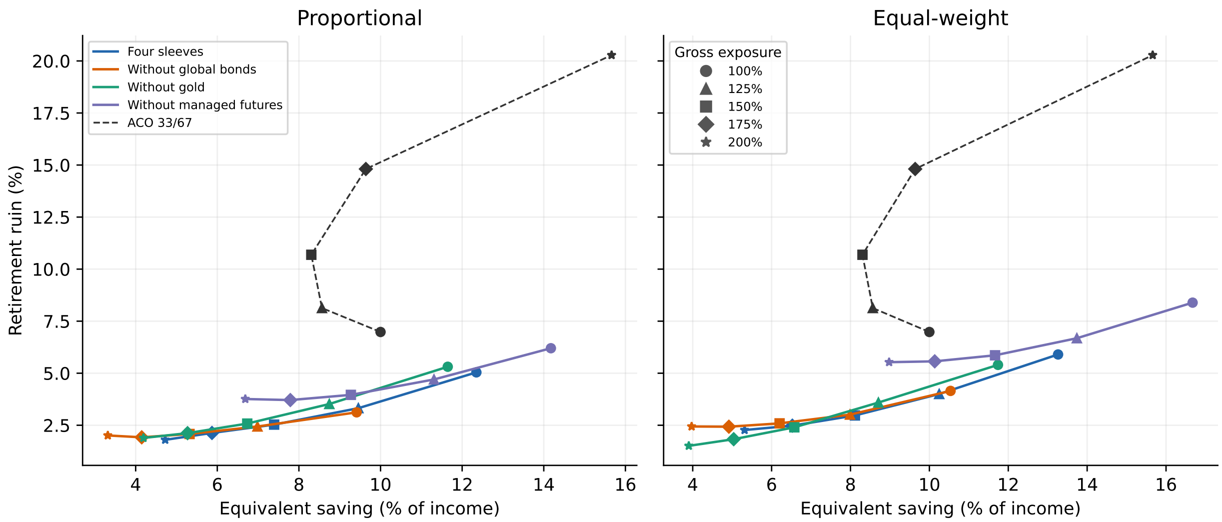

Figure 1: Saving and ruin across complete and ablated portfolio ladders. Marker shapes identify gross exposure: 100, 125, 150, 175, and 200%. Each curve uses fixed sleeve proportions; removing a sleeve reallocates its weight pro rata. ACO is shown at the same five exposures. All 45 strategies share 10,000 paths from the raw 1927–2025 panel and the unlevered ACO utility target. The source-screened counterpart is in Appendix I.

4.4 Capital preservation without portfolio borrowing

At 100% gross exposure, both diversified families improve capital preservation without a portfolio-level loan. The equal-weight rule has 15.09% annual volatility and 5.90% ruin, against 25.28% and 6.98% for ACO. The proportional rule has 15.66% volatility and 5.04% ruin. These volatilities are about 60% of the benchmark's full-panel value. With $G = 1$, the financing-spread term $\max(G - 1, 0)\phi$ is zero, so these results do not depend on that spread. Fees, hedge costs and the derivative exposures inside managed futures remain.

The utility comparison differs. Equivalent saving rises to 13.26% for the equal-weight rule and 12.35% for the proportional rule, against 10% for ACO. Lower ruin therefore does not imply higher expected utility: the latter also values retirement consumption and bequests. In these fixed families, capital preservation improves before borrowing; additional exposure is needed to reach benchmark utility at the same saving rate.

4.5 Financing a common planned retirement income

A complementary common-withdrawal test fixes each household's planned real draw at 4% of its paired unlevered ACO retirement wealth. Relative to a candidate portfolio's own retirement capital, this draw can exceed 4% substantially: for equal-weight 175%, its median implied rate is 2.93%, but the 90th percentile is 6.83%. This test jointly evaluates accumulation and the capacity to finance the benchmark's planned income; it differs from withdrawing 4% of each strategy's own capital (Appendix F.1). Using the main experiment's denominator of all simulated lifetimes, the proportional 175% rule has a 2.17% probability of failing to fund that draw, compared with 6.98%

for ACO and 7.98% for equal-weight 175%. ACO's 698 failures reproduce the main experiment exactly. Conditional on reaching retirement with a positive target, the respective rates are 2.21%, 7.10%, and 8.12%; Appendix F.1 reconciles the denominators and reports the path-by-path identity check. The equal-weight rule requires withdrawals above 4% of its own capital on 32.26% of eligible paths. Its less favorable result on this criterion coexists with lower ruin under its own 4% withdrawal rule and a lower utility-equivalent saving rate. The proportional family improves the common income comparison as well, indicating that the broader diversification benefit can survive this more demanding comparison of accumulation and retirement funding. Annual volatility alone does not explain the difference.

4.6 Historical windows and concentration of variance

The source screen removes 33 resident observations and applies the same market-year exclusions before constructing international equities and global bonds, leaving 1,528 country-years. It combines the documented disruption list, the annual inflation screen, and Italy 1942 identified by the influence audit. Italy 1942 is not removed by the original inflation or closure rules. The resulting screen is a retrospective sample convention, not evidence that all retained returns were tradable. In particular, removing crisis states changes the economic distribution as well as the measurement problem. Table 3 therefore shows the complete raw and screened comparisons side by side. At 175%, screened equal-weight volatility is 15.17%, against 17.02% for ACO; ruin is 0.57% against 5.04%, and equivalent saving is 5.93% against 10%. Clipping each risky sleeve at the pooled 0.5th and 99.5th percentiles instead produces volatilities of 15.47% and 17.03%, ruin of 1.58% and 7.35%, and saving of 6.17% and 10%. The two treatments both support the ordering, but their different ruin levels show why winsorization and sample exclusion are distinct experiments.

We retain 1927–2025 as the primary sample. Large currency movements in the JST observations for Italy 1942 and France 1946 propagate into converted foreign returns. We retain these observations and measure their influence through separate source-event exclusions.

Italy 1942 accounts for 48.35% of ACO's sum of squared deviations from its full-panel mean, and 52.25% for the equal-weight 175% rule. Deleting this common row solely as a diagnostic reduces their volatilities to 18.17% and 17.65%, from 25.28% and 25.54%. Their respective top-five concentration shares are 54.61% and 64.10%; those sets need not contain identical rows. The full-panel volatility comparison is therefore strongly influenced by these observations, whereas the ruin and utility comparisons favor the diversified rule both with and without them.

Table 4 applies the two source-event influence diagnostics consistently. A source event is removed from the resident bootstrap row and from every reconstructed international-equity and global-bond basket for that year. Under the Italy-1942 event removal, the equal-weight 175% rule has 2.42% ruin and a 6.41% equivalent saving rate, versus 7.19% and 10% for ACO. Its annual volatility is 17.64%, against 18.17% for ACO. Removing France 1946 as well changes the levels but not the direction of the comparison. In both diagnostics, the diversified rule combines lower volatility, lower ruin, and a lower saving requirement.

Table 4: Source-event influence diagnostics

Diagnostic	Portfolio	Volatility	Ruin	Equivalent saving
Italy 1942 event	ACO	18.17%	7.19%	10.00%
	ACO 175%	31.26%	15.79%	10.02%
	Equal-weight 175%	17.64%	2.42%	6.41%
Italy 1942; France 1946 events	ACO	17.50%	7.17%	10.00%
	ACO 175%	29.90%	15.21%	10.28%
	Equal-weight 175%	16.00%	2.53%	6.41%

10,000 paired simulations per diagnostic. The corresponding 95% paired Monte Carlo intervals for the equal-weight ruin difference are $[-5.25, -4.29]$ and $[-5.12, -4.16]$ percentage points. This table can be reproduced using the replication commands in the source provenance log.

The two later windows in Table 5 remove entire early regimes. The windows and source-influence diagnostics were selected after examining the full-panel results and serve as historical sensitivities. The fixed 200% rules retain lower ruin and equivalent saving in each window, while their volatility relative to ACO changes.

Table 5: Fixed allocations across historical windows

Window	Portfolio	Volatility	Ruin	Equivalent saving
1927–2025	ACO	25.28%	6.98%	10.00%
1927–2025	ACO 200%	50.28%	20.28%	15.66%
1927–2025	Proportional 200%	30.31%	1.80%	4.72%
1927–2025	Equal-weight 200%	29.12%	2.27%	5.31%
1950–2025	ACO	16.59%	5.42%	10.00%
1950–2025	ACO 200%	32.71%	18.85%	18.09%
1950–2025	Proportional 200%	16.95%	0.29%	5.01%
1950–2025	Equal-weight 200%	16.54%	0.40%	6.01%
1970–2025	ACO	17.24%	6.67%	10.00%
1970–2025	ACO 200%	34.00%	23.67%	29.86%
1970–2025	Proportional 200%	17.13%	0.07%	3.23%
1970–2025	Equal-weight 200%	16.90%	0.06%	3.22%

10,000 paired simulations per window; 1,561, 1,213 and 894 country-years, respectively. Fixed-notional hedges and baseline costs throughout.

Post-1970 ablations make the regime dependence more explicit (Appendix K). At 175%, the complete proportional and equal-weight families each require 4.16% equivalent saving, with estimated ruin of 0.04% and 0.03%. Removing gold raises proportional ruin to 0.64% and required saving to 4.63%; removing it from equal weight raises ruin to 0.08% and saving to 4.19%, while reducing annual volatility. Removing bonds raises volatility in both families and ruin to 0.20% and 0.16%, although required saving falls. Removing MF raises ruin to 0.68% and 0.81%. Gold and bonds thus contribute to capital preservation in this later window, with different utility and volatility trade-offs. Very small ruin estimates represent few simulated events and should not be interpreted as precise distinctions between families. These window comparisons change several regimes jointly and do not isolate a causal effect of ending administered gold prices.

Table 6: Implementation checks

Specification	ACO ruin	Proportional 200% ruin	Equal-weight 200% ruin
Corrected baseline	6.98%	1.80%	2.27%
Legacy bootstrap end treatment	7.38%	1.84%	2.26%
Ideal full-value hedge	6.98%	1.82%	2.27%
Frozen legacy return screen (diagnostic only)	7.38%	1.57%	2.02%
Leave-one-source-country-out range	7.10–8.66%	0.90–2.88%	1.36–3.16%

Note: First four rows use 10,000 paired simulations; the leave-one-out range uses 5,000 per omission. The 200% columns assess robustness of the portfolio ordering; their volatility exceeds that of ACO in the baseline.

The outer calendar-history experiment also supports a broader, rather than composition-invariant, conclusion. Proportional 175% has lower estimated ruin than ACO in 98–100% of outer histories across the three block lengths; equal-weight 175% does so in 86–97%. With ten-year outer blocks, their median ruin differences are respectively -5.15 and -4.70 percentage points. The 5th–95th percentile ranges are $[-12.90, -1.19]$ and $[-13.11, 0.11]$. Equivalent-saving ranges are 3.90–8.40% and 3.51–11.33%. Hence the proportional family's advantage is more stable in this diagnostic; the equal-weight ordering can reverse in some resampled histories (Appendix D).

4.7 Exposure and implementation sensitivity

Increasing exposure to 200% further reduces ruin and equivalent saving. The proportional rule reaches 1.80% ruin and 4.72% equivalent saving; the equal-weight rule reaches 2.27% and 5.31%. Their full-panel annual volatilities rise to 30.31% and 29.12%, respectively, compared with 25.28% for ACO. Higher exposure thus improves the lifecycle outcomes while increasing annual return dispersion.

Table 6 records the checks most relevant to the construction. A return-dependent screen lowers measured volatility substantially. We report it as a construction diagnostic alongside the unscreened baseline. Reverting to the former bootstrap end treatment or to the idealized hedge changes levels modestly but does not reverse the ranking. A leave-one-source-country-out exercise removes the country from resident observations and from the global equity and bond baskets. Across the 16 omissions, benchmark ruin ranges from 7.10% to 8.66%, while the two 200% sensitivity endpoints remain below it (ranges 0.90–2.88% and 1.36–3.16%). This supports robustness of the ordering, while also showing that its absolute risk depends on the historical sample.

At a 100-basis-point financing spread, the 200% proportional and equal-weight sensitivity rules have ruin of 2.44% and 3.16%, respectively; at 200 basis points this rises to 3.82% and 4.91%. Higher financing costs erode the gains, although both rules retain lower ruin than the 6.98% equity benchmark at these spreads.[3]

The new cost and proxy-return controls focus on the central 175% exposure (Appendix L). At a 100 bp financing spread, the proportional and equal-weight rules require 6.78% and 7.53% saving

[3]Complete numerical outputs, replication scripts, and scope limitations regarding synthetic series are archived in the online repository and detailed in Appendix G.

and have 2.51% and 3.24% ruin. Relative to efficient financing through futures-based funds or short index-option boxes, this 100 bp scenario is already costly; it is an adverse-cost control above the 30 bp baseline. A quoted retail loan rate must be compared with the contemporaneous bill rate used in the model: its premium over that bill need not equal the broker's markup over its own reference benchmark. A financing spread of 200 bp, additional to the resident bill, leaves saving requirements at 8.35% and 9.29% for the proportional and equal-weight families. At 250 bp these become 9.28% and 10.36%; at 300 bp, 10.34% and 11.58%. Thus the tested grid brackets loss of the saving advantage at 250–300 bp and 200–250 bp, respectively. At 300 bp their ruin remains 5.32% and 6.92%, versus 6.98% for unlevered ACO; the latter difference is small. These persistent high spreads are severe stresses for efficient futures-based financing, whose cost distinction from retail margin borrowing is discussed in Section 3.3. They are not total borrowing rates.

Separately, deducting an additional 300 bp annually from the net MF sleeve leaves required saving at 7.47% and 9.39%, with ruin of 3.01% and 4.76%. The equal-weight saving advantage disappears between 300 and 400 bp of MF return deduction, whereas the proportional rule retains it at 600 bp and loses it at 700 bp. These are additive annual-return stress tests after baseline fees and costs, preserving the proxy's return fluctuations. They establish tolerance to lower returns without validating omitted carry, trading constraints, or a different crisis-response profile.

5 Individual allocation, market capacity, and aggregate adoption

Our simulations evaluate a household whose trades do not move market prices. Historical returns are held fixed when portfolio weights change. This is a partial-equilibrium comparison: it measures household outcomes under a given return environment, without solving for how widespread adoption would change that environment. The distinction applies to the equity benchmark as well as the multi-asset extensions.

For equities, Gabaix and Koijen (2023) develop an inelastic-markets framework in which institutional allocation constraints limit the response of demand to prices, allowing portfolio flows to move aggregate valuations substantially. The mechanism is not simply a fixed number of shares: it also concerns how willing existing investors are to adjust their holdings. Applied to our setting, this implies that a broad retirement reallocation toward equities cannot be assumed to preserve historical returns. Buying existing shares transfers ownership; it does not automatically create productive capital. With expected cash flows held fixed, a higher purchase price lowers the prospective return.

For gold, the mechanism is straightforward: if retirement funds seek to buy much more of it, its price must rise enough to encourage existing owners to sell or producers to supply more. Erb and Harvey (2013) discuss this demand effect and find that high inflation-adjusted gold prices have historically been followed by lower real returns. Investors who already own gold benefit from the price rise; those buying afterwards pay more for the same asset. Large-scale adoption could therefore change the returns available to future investors, even if gold continues to diversify their portfolios.

For trend following, the concern is that many funds may buy or sell the same contracts at the same time. Larger orders can become more expensive to execute and can themselves move prices. Nick Baltas (2019) finds weaker subsequent performance when many investors hold similar momentum positions. His evidence concerns strategies that rank assets against one another, so it does not directly measure the capacity of our rule, which uses each asset's own past returns. Nick Baltas and Kosowski (2013) find no statistically significant evidence that scale constrained performance in their historical CTA sample. This supports feasibility at the scale they studied, without establishing how much additional capital the strategy could absorb. That depends on the contracts traded, the size and frequency of orders, and the liquidity available to execute them.

This distinction matters equally for interpreting ACO. Their household simulations also resample historical returns without making prices respond to widespread adoption of the recommended equity allocation (Anarkulova, Cederburg, and O'Doherty, 2025). Neither study therefore establishes what would happen if the retirement system as a whole adopted its portfolios. Our finding is that broader diversification improves household outcomes under the same historical-return approach used to evaluate the equity benchmark. The cost and return stresses show how much those gains can withstand higher costs or lower MF returns. Assessing a system-wide reallocation would require an additional analysis of how investor demand changes prices and trading conditions for all the portfolios compared.

For an individual investor whose trades are too small to move market prices, these aggregate considerations do not by themselves overturn the portfolio comparisons reported here. They concern how widespread adoption could change future returns, rather than the market impact of implementing the allocation at household scale.

Conclusion

The matched-exposure comparison separates diversification from borrowing. Levering ACO to 175% produces 14.81% simulated ruin, versus 2.13% and 2.48% for the diversified families, with higher saving requirements. The advantage of broader asset-class exposure therefore remains when the equity benchmark uses the same amount of portfolio borrowing.

Broadening the investment universe changes the comparison of fixed lifecycle allocations. Our annual public-data implementation recovers the advantage of international over domestic equities. Combining global bonds, gold, and managed futures then improves retirement outcomes: the equal-weight portfolio at 175% exposure reduces simulated ruin from 6.98% to 2.48% and matches benchmark utility with a saving rate of 6.53% rather than 10%.

Diversification and exposure make distinct contributions. At 100% exposure, both diversified families reduce volatility and ruin without portfolio-level borrowing. Additional exposure raises retirement resources and brings the saving requirement below that of the equity benchmark. Under both source-event influence diagnostics, the equal-weight 175% portfolio also has lower annual volatility than ACO, alongside lower ruin and equivalent saving. The later historical windows preserve the favorable lifecycle ranking. At 175% exposure, all six three-sleeve ablations retain lower ruin than unlevered ACO, and five also require less saving to match its utility. The gains thus extend across several diversified compositions, with managed futures making the largest contribution to capital preservation in the ablation comparisons.

The additional evidence sharpens this interpretation. Post-1970 ablations show a capital-preservation role for gold and bonds, while MF-return and financing stresses quantify how far the gains can erode before benchmark utility requires 10% saving again. Historical-panel resampling and a common ACO income target favor the proportional family more consistently than equal weighting. The latter's weaker common-income outcome remains a limitation, even though its own-capital ruin and baseline utility improve. The result is therefore an advantage for several diversified allocations under stated conditions, rather than universal dominance by one fixed rule.

The economic implication is that the choice of asset classes and the choice of total exposure belong together in lifecycle portfolio design. In the allocations studied here, international equity diversification provides a strong starting point; diversification across asset classes improves capital preservation, and scaling that diversified portfolio improves the financing of retirement.

References

Anarkulova, Aizhan, Scott Cederburg, and Michael S. O'Doherty (2025). "Beyond the Status Quo: A Critical Assessment of Lifecycle Investment Advice". Working paper, July 10 revision. DOI: 10.2139/ssrn.4590406.

Asness, Clifford S. (1996). "Why Not 100% Equities". In: *Journal of Portfolio Management* 22.2, pp. 29–34. DOI: 10.3905/jpm.1996.29.

Baltas, Nick (2019). "The Impact of Crowding in Alternative Risk Premia Investing". In: *Financial Analysts Journal* 75.3, pp. 89–104. DOI: 10.1080/0015198X.2019.1600955.

Baltas, Nick and Robert Kosowski (2013). "Momentum Strategies in Futures Markets and Trend-following Funds". Working paper, January 5 version. DOI: 10.2139/ssrn.1968996.

— (2020). "Demystifying Time-Series Momentum Strategies: Volatility Estimators, Trading Rules and Pairwise Correlations". In: *Market Momentum: Theory and Practice*. Wiley; earlier circulated as a 2013 working paper, pp. 30–62.

Baltas, Nikitas (2015). "Trend-Following, Risk-Parity and the Influence of Correlations". In: *Risk-Based and Factor Investing*. Elsevier.

Bengen, William P. (1994). "Determining Withdrawal Rates Using Historical Data". In: *Journal of Financial Planning* 7, pp. 171–180.

Board of Governors of the Federal Reserve System (2026). *Foreign Exchange Rates—H.10 Release*. https://www.federalreserve.gov/releases/h10/. Daily dollar exchange-rate observations; the last published observation of each month is used for the managed-futures currency sleeve and USD conversion.

Campbell, John Y. and Luis M. Viceira (2002). *Strategic Asset Allocation: Portfolio Choice for Long-Term Investors*. Oxford: Oxford University Press.

Clare, Andrew, James Seaton, Peter N. Smith, and Stephen Thomas (2017). "Reducing Sequence Risk Using Trend Following and the CAPE Ratio". In: *Financial Analysts Journal* 73.4, pp. 91–103.

— (2019). *Can Sustainable Withdrawal Rates be Enhanced by Trend Following?* SSRN Working Paper No. 3019089.

CME Group (2024). *Index Options Box Spreads as Financing Tool*. URL: https://www.cmegroup.com/articles/2024/index-options-box-spreads-as-financing-tool.html (visited on 09/05/2026).

Cocco, João F., Francisco J. Gomes, and Pascal J. Maenhout (2005). "Consumption and Portfolio Choice over the Life Cycle". In: *Review of Financial Studies* 18, pp. 491–533.

De Nardi, Mariacristina, Eric French, and John B. Jones (2010). "Why Do the Elderly Save? The Role of Medical Expenses". In: *Journal of Political Economy* 118, pp. 39–75.

DeMiguel, Victor, Lorenzo Garlappi, and Raman Uppal (2009). "Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy?" In: *Review of Financial Studies* 22.5, pp. 1915–1953. DOI: 10.1093/rfs/hhm075.

Erb, Claude B. and Campbell R. Harvey (2013). “The Golden Dilemma”. In: *Financial Analysts Journal* 69.4, pp. 10–42.

Estrada, Javier (2018). “Maximum Withdrawal Rates: An Empirical and Global Perspective”. In: *Journal of Retirement* 5.3, pp. 57–71.

Federal Reserve Bank of St. Louis (2026). *Federal Reserve Economic Data*. https://fred.stlouisfed.org/. Access point for the monthly OECD, NBER, Bank of England, and H.10 source series.

Gabaix, Xavier and Ralph S. J. Koijen (2023). “In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis”. Working paper, December 23 revision; Swiss Finance Institute Research Paper No. 20-91. DOI: 10.2139/ssrn.3686935.

Guvenen, Fatih, Fatih Karahan, Serdar Ozkan, and Jae Song (2021). “What Do Data on Millions of U.S. Workers Reveal about Lifecycle Earnings Dynamics?” In: *Econometrica* 89, pp. 2303–2339.

Hurst, Brian, Yao Hua Ooi, and Lasse Heje Pedersen (2017). “A Century of Evidence on Trend-Following Investing”. In: *Journal of Portfolio Management* 44.1, pp. 15–29.

Interactive Brokers (2026). *Margin Rates and Financing*. Published tier spreads use the broker reference benchmark, not a one-year Treasury yield. URL: https://www.interactivebrokers.com/en/trading/pricing-margin-rates.php (visited on 09/05/2026).

Jordà, Òscar, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick, and Alan M. Taylor (2019). “The Rate of Return on Everything, 1870–2015”. In: *Quarterly Journal of Economics* 134, pp. 1225–1298.

— (2023). *The Jordà–Schularick–Taylor Macrohistory Database, Release 6*. https://www.macrohistory.net/database/. Annual panel of 18 advanced economies, 1870–2020; series `eq_tr`, `bond_tr`, `bill_rate`, `cpi`, exchange rates, and real GDP.

Kuvshinov, Dmitry and Kaspar Zimmermann (2022). “The Big Bang: Stock Market Capitalization in the Long Run”. In: *Journal of Financial Economics* 145.2, pp. 527–552.

Lempérière, Y., C. Deremble, P. Seager, M. Potters, and J.-P. Bouchaud (2014). “Two Centuries of Trend Following”. In: *Journal of Investment Strategies* 3.3, pp. 41–61.

London Bullion Market Association (2025). *LBMA Gold Price, afternoon (PM) fixing*. https://www.lbma.org.uk/prices-and-data/precious-metal-prices. U.S. dollar per troy ounce; used from 1968.

Markowitz, Harry (1952). “Portfolio Selection”. In: *Journal of Finance* 7, pp. 77–91.

Moskowitz, Tobias J., Yao Hua Ooi, and Lasse Heje Pedersen (2012). “Time Series Momentum”. In: *Journal of Financial Economics* 104, pp. 228–250.

Müller, Karsten, Chenzi Xu, Mohamed Lehbib, and Ziliang Chen (2025). *The Global Macro Database: A New International Macroeconomic Dataset*. Working Paper 33714. Used for 2021–2025 inflation and short rates; https://www.globalmacrodata.com/. National Bureau of Economic Research.

National Bureau of Economic Research (2016). *NBER Macrohistory Database*. Distributed via FRED, Federal Reserve Bank of St. Louis. Chicago and New York wholesale commodity prices, 1841–1956, for the managed-futures proxy.

Officer, Lawrence H. and Samuel H. Williamson (2024). *The Price of Gold, 1257–Present, and Exchange Rates Between the United States Dollar and Forty-one Currencies*. MeasuringWorth. https://www.measuringworth.com/. Gold: annual U.S. dollar price through 1967; exchange rates used in the managed-futures currency sleeve.

Organisation for Economic Co-operation and Development (2026). *Main Economic Indicators*. Distributed through Federal Reserve Economic Data (FRED). Monthly national equity-price indexes, sovereign yields, and short rates used in the managed-futures input snapshot.

Poterba, James M. and Lawrence H. Summers (1988). "Mean Reversion in Stock Prices: Evidence and Implications". In: *Journal of Financial Economics* 22, pp. 27–59.

Return Stacked Portfolio Solutions (Nov. 7, 2024). *Return Stacking and the Cost of Leverage*. Industry research on implied financing in equity-index and Treasury futures. URL: https://www.returnstacked.com/return-stacking-and-the-cost-of-leverage/ (visited on 09/05/2026).

Social Security Administration (2020). *Actuarial Life Table*. Office of the Chief Actuary.

Tidal Trust II (May 31, 2024). *Return Stacked ETFs: Statement of Additional Information*. URL: https://www.returnstackedetfs.com/wp-content/uploads/2024/06/rsbt-sai_053024_Web.pdf (visited on 09/05/2026).

— (Apr. 27, 2026). *Return Stacked Global Stocks and Bonds ETF: Summary Prospectus*. URL: https://www.sec.gov/Archives/edgar/data/1924868/000199937126009149/rssb-497k_042726.htm (visited on 09/05/2026).

Wang, Joseph (Mar. 26, 2025). *Long-Dated Box Spreads: A Better Way to Buy a Home*. Practitioner contribution by a SyntheticFi cofounder; market estimates dated February 24, 2025. Cboe, Guest Author Series. URL: https://www.cboe.com/insights/posts/long-dated-box-spreads-a-better-way-to-buy-a-home-updated/ (visited on 09/05/2026).

World Bank (2025). *Commodity Markets "Pink Sheet" Monthly Prices*. https://www.worldbank.org/en/research/commodity-markets. Twenty-one monthly commodity series used in the managed-futures proxy from 1960.

A Asset-Class Return Histories

The cumulative-return exhibit reports annual U.S.-resident returns under the current fixed-notional hedge. The logarithmic axis makes proportional changes comparable across dates.

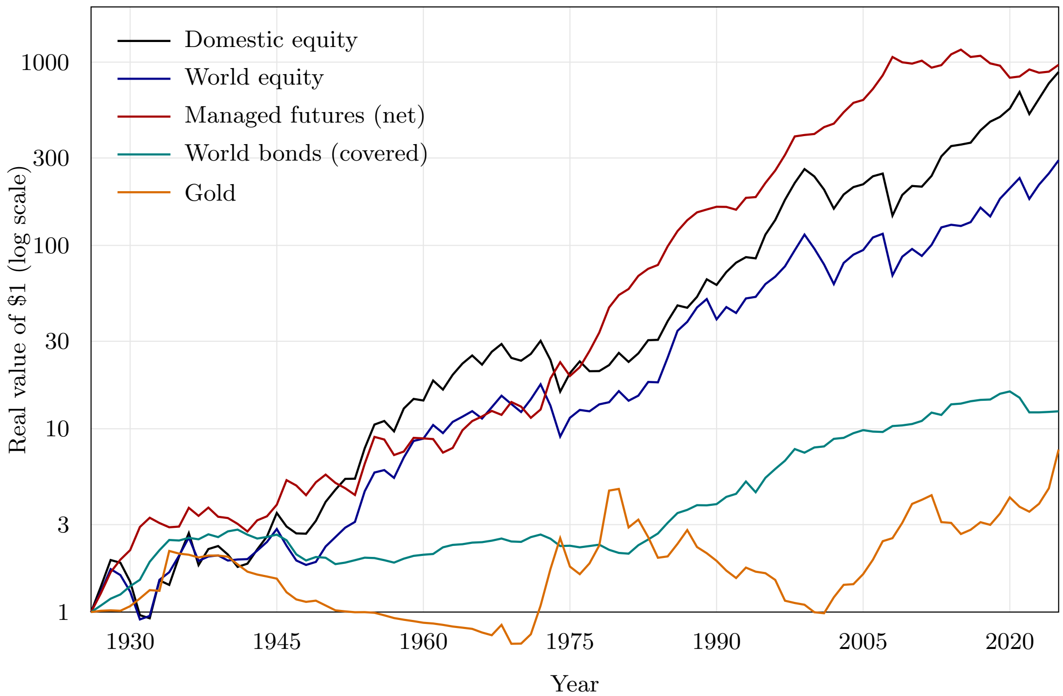

Figure 2: Cumulative real return by asset class, U.S. resident, 1927–2025. Managed futures are the reconstructed time-series-momentum proxy net of the 0.85% fee and 0.61% turnover cost. World bonds are the covered equal-weight basket of all panel sovereigns. Gold is the spliced MeasuringWorth–LBMA price less custody. Equity and gold are unhedged, while bonds and managed futures are currency-hedged under covered interest parity. The managed-futures proxy compounds at close to the equity rate over the century, but its decade-by-decade path is far more volatile before 1960, when the underlying market set is thin.

B Properties of the reconstructed managed-futures proxy

This appendix documents the managed-futures rule and compares its returns with commercial indexes and funds. The rule combines momentum signals at several horizons and scales positions using lagged volatility. The economic motivation follows Moskowitz, Ooi, and Pedersen (2012) and Hurst, Ooi, and Pedersen (2017). The external comparisons measure how closely the proxy captures the return variation of commercial trend-following strategies over their common history.

B.1 Monthly inputs and the annual simulation series

The annual lifecycle panel and the monthly signal inputs are distinct data layers. The monthly snapshot supplies the managed-futures construction; complete calendar years are then compounded for the annual simulation. The source table records the original reconstruction's inputs and transformations. Price-only equities, synthetic bond returns, omitted commodity carry and roll, and incomplete early currency coverage limit the economic interpretation of the reconstructed return.

The synthetic bond inputs are excess P&Ls rather than foreign bond holdings. The construction converts those P&Ls at month end without adding a foreign principal: a zero local excess return is therefore not turned into an exchange- rate return. Price-index equities remain cash assets and retain their principal currency exposure.

The proxy includes the U.S.-dollar collateral return recorded in the monthly snapshot. For the annual household simulation, its dollar net asset value (NAV) is treated as a foreign asset. The baseline applies the initial-notional forward in equation (4), using the recorded collateral return for the foreign cash leg, as it does for global bonds. Exchange-rate exposure on subsequent NAV gains or losses remains.

The alternative full-value convention replaces foreign collateral with the resident bill for the entire terminal asset value. A futures strategy collateralized directly in the household's currency would be a different implementation.

Table 7: Monthly data used to construct the managed-futures proxy

Input	Monthly source and transformation	What enters the MF rule
Equities	National price indexes from OECD Main Economic Indicators via FRED, and before their availability, the U.S., U.K., German, and French NBER Macrohistory monthly price series.	18 local-currency price returns. Dividends are absent.
Rates	Monthly sovereign yields from OECD via FRED, augmented before their start by NBER U.S. yields, U.K. Consol/Bank of England yields, and the French 3% rente. Each yield pair is priced as a par ten-year bond, then the lagged three-month cash return is subtracted.	18 synthetic local-currency bond excess returns, not changes in yields and not observed futures returns.
Commodities	NBER/BLS monthly wholesale prices for the early history and the World Bank Pink Sheet from 1960. Gold uses the official U.S. price before 1960 and the Pink Sheet thereafter, and silver starts in 1960.	19 commodity spot-price returns plus gold and silver. Carry, basis, and futures roll are unavailable and therefore set to zero.
Currencies	Board of Governors H.10 daily dollar quotations via FRED, reduced to the last available observation in each month, with a lagged three-month interest differential forming a synthetic forward excess return.	AUD, CAD, CHF, GBP, JPY from 1971 and EUR from 1999. Earlier European currencies are not backfilled.
Collateral and deflator	Monthly three-month cash rates from OECD via FRED, with a lagged prior-year JST/GMD fallback only where monthly rates do not exist, plus monthly U.S. CPI for deflation.	U.S. cash collateral is added once to the USD MF NAV, and CPI converts the final monthly USD return to real terms.

Note: The machine-readable source-segment file records the provider, series identifier, dates, transformation, and source seam for every market. The monthly snapshot is frozen for this replication. The first month of each source segment receives no return, preventing differences in source price levels from entering the trading signal.

B.2 Comparison with commercial trend-following indexes

These correlations compare monthly dollar returns independently of the resident panel, annual hedge, and lifecycle bootstrap. The external comparison places the net monthly proxy alongside SG CTA, SG Trend, Barclay BTOP50, and the KMLMSIM simulation over their common period. The indexes are external benchmarks and enter no lifecycle resampling. Their proprietary inputs remain subject to their source terms.

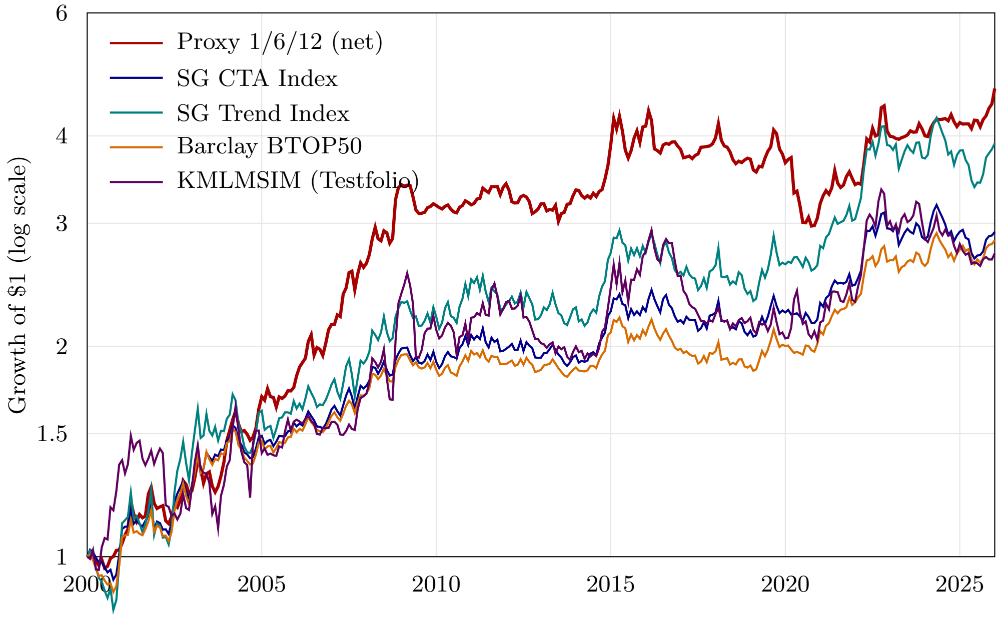

Figure 3: Growth of one dollar, monthly, 2000–2025: the net managed-futures proxy (variant 1/6/12, the one used in the paper) against three commercial trend-following indexes and KMLMSIM, the Testfolio simulation of KMLM. SG CTA, SG Trend, and BTOP50 are net of index fees, and the proxy is net of the 0.85% management fee and the monthly measured trading costs. KMLMSIM uses the KFA MLM Index less 0.90% per year before the ETF's inception. The SG and BarclayHedge indexes are proprietary and are used here only under the terms held for this research.

Table 8: Monthly correlation of the net proxy with trend-following benchmarks

Benchmark	Corr.	Common window
SG CTA Index	0.57	2000-01–2025-12 (312 mo.)
SG Trend Index	0.55	2000-01–2025-12 (312 mo.)
Barclay BTOP50 Index	0.59	2000-01–2025-12 (312 mo.)
KMLMSIM (Testfolio simulation)	0.53	2000-01–2025-12 (312 mo.)
DBMF ETF (live)	0.61	2019-06–2025-12 (79 mo.)
KMLM ETF (live)	0.63	2021-01–2025-12 (60 mo.)

Memo, 2000–2025: the three commercial indexes correlate 0.96–0.97 with each other, whereas KMLMSIM correlates 0.60 with SG CTA. The proxy near 0.51 sits between these two reference patterns.

Note: contemporaneous Pearson correlation of monthly returns over the common months shown. KMLMSIM is Testfolio's KFA MLM Index-based simulation of KMLM before the ETF's inception, and the "live" rows use the ETFs' own return history from the first full month after listing.

B.3 The proxy among commercial benchmarks and investable funds

The broader comparison reports the updated pairwise correlation matrix, the matrix on a common window, and the regressions on the proxy. Pairwise windows differ when funds have different inception dates. The common-window matrix addresses that comparability issue, at the cost of a shorter history. The regressions report exposures to the proxy, intercepts, and tracking errors, describing both common return variation and differences across products.

Table 9: Correlation matrix of monthly returns: the net proxy, commercial benchmarks, and investable managed-futures funds

	SG CTA	SG Trend	BTOP50	KMLMSIM	DBMF	KMLM	QMHIX	AHLIX	AHLT	IMF	ISMF	WTMF	APEX
Proxy	**0.57**	**0.55**	**0.59**	**0.53**	**0.61**	**0.63**	**0.53**	**0.56**	**0.68**	**0.68**	**0.62**	**0.19**	**0.18**
SG CTA	—	0.97	0.97	0.60	0.86	0.77	0.83	0.84	0.86	0.76	0.87	0.40	0.50
SG Trend	0.97	—	0.96	0.60	0.86	0.74	0.85	0.86	0.85	0.72	0.89	0.42	0.46
BTOP50	0.97	0.96	—	0.58	0.85	0.73	0.79	0.83	0.84	0.69	0.82	0.41	0.57
KMLMSIM	0.60	0.60	0.58	—	0.66	1.00	0.75	0.63	0.69	0.89	0.86	0.34	0.33
DBMF	0.86	0.86	0.85	0.66	—	0.68	0.73	0.74	0.77	0.29	0.53	0.24	0.50
KMLM	0.77	0.74	0.73	1.00	0.68	—	0.74	0.65	0.69	0.89	0.86	0.12	0.33
QMHIX	0.83	0.85	0.79	0.75	0.73	0.74	—	0.70	0.74	0.54	0.82	0.33	0.40
AHLIX	0.84	0.86	0.83	0.63	0.74	0.65	0.70	—	0.85	0.76	0.83	0.39	0.28
AHLT	0.86	0.85	0.84	0.69	0.77	0.69	0.74	0.85	—	0.82	0.90	0.13	0.56
IMF	0.76	0.72	0.69	0.89	0.29	0.89	0.54	0.76	0.82	—	0.86	0.28	0.40
ISMF	0.87	0.89	0.82	0.86	0.53	0.86	0.82	0.83	0.90	0.86	—	0.42	0.59
WTMF	0.40	0.42	0.41	0.34	0.24	0.12	0.33	0.39	0.13	0.28	0.42	—	0.01
APEX	0.50	0.46	0.57	0.33	0.50	0.33	0.40	0.28	0.56	0.40	0.59	0.01	—

Note: contemporaneous Pearson correlation of monthly returns, each pair over the overlap of its two histories within 2000-01 to 2025-12, the frame of the appendix. The first month in that frame is 2000-01 for the SG indexes (their full history), 2019-06 for DBMF, 2021-01 for KMLM, 2011-02 for WTMF, 2013-08 for QMHIX, 2014-09 for AHLIX, 2022-04 for Apex, 2023-10 for AHLT, and 2025-04 for IMF and ISMF; BTOP50 (from 1987) and KMLMSIM (from 1988) are restricted to the common frame, and the proxy is available from 1926-03, so the number of overlapping months of a pair is determined by the later of its two start months. The index returns are the proprietary SG and BarclayHedge series and the testfol.io series of Table 8; fund returns are total returns net of fund fees from Yahoo Finance adjusted closes. Shading is proportional to the correlation; the horizontal rule separates the benchmark block from the investable funds. QMHIX, AHLIX, AHLT, IMF, ISMF, DBMF, and KMLM are pure trend-following products; WTMF is a systematic equity-timing strategy and Apex a multi-strategy fund.

Table 10: Correlation matrix over a common window: the proxy and the benchmarks and funds with at least ten years of history

	SG CTA	SG Trend	BTOP50	KMLMSIM	WTMF	QMHIX	AHLIX
Proxy	**0.58**	**0.59**	**0.62**	**0.54**	**0.20**	**0.53**	**0.56**
SG CTA	—	0.98	0.97	0.72	0.39	0.84	0.84
SG Trend	0.98	—	0.96	0.72	0.40	0.86	0.86
BTOP50	0.97	0.96	—	0.69	0.41	0.79	0.83
KMLMSIM	0.72	0.72	0.69	—	0.31	0.75	0.63
WTMF	0.39	0.40	0.41	0.31	—	0.33	0.39
QMHIX	0.84	0.86	0.79	0.75	0.33	—	0.70
AHLIX	0.84	0.86	0.83	0.63	0.39	0.70	—

Note: contemporaneous Pearson correlation of monthly returns, every pair evaluated over the same window 2014-09 to 2025-12 (136 months), the maximal common window of the eight series (proxy from 1926-03; SG indexes from 2000-01, BTOP50 and KMLMSIM restricted to the common frame; WTMF from 2011-02, QMHIX from 2013-08, AHLIX from 2014-09). DBMF (79 months) and KMLM (60 months) do not meet the ten-year rule. Sources and conventions as in Table 9.

Table 11: Regressions of managed-futures benchmarks and investable funds on the net proxy

Series	n	α (%/mo.)	β	R^2	σ_f/σ_p	TE (%/yr.)
SG CTA	312	+0.12 (+1.0)	0.47 (+12.1)	0.32	0.83	9.1
SG Trend	312	+0.13 (+0.7)	0.70 (+11.5)	0.30	1.27	11.6
BTOP50	312	+0.12 (+1.1)	0.44 (+12.7)	0.34	0.76	8.7
KMLMSIM	312	−0.01 (−0.0)	0.79 (+11.1)	0.28	1.48	13.4
DBMF	79	+0.49 (+1.7)	0.63 (+6.8)	0.38	1.02	9.7
KMLM	60	−0.11 (−0.3)	0.82 (+6.1)	0.39	1.31	10.1
QMHIX	149	+0.25 (+0.8)	0.76 (+7.5)	0.28	1.45	13.2
AHLIX	136	+0.26 (+1.3)	0.49 (+7.9)	0.32	0.88	9.7
AHLT	27	−0.62 (−0.9)	1.58 (+4.6)	0.46	2.33	11.8
IMF	9	−2.29 (−1.7)	1.31 (+2.4)	0.46	1.93	11.5
ISMF	9	+0.48 (+0.8)	0.51 (+2.1)	0.39	0.83	6.4
WTMF	179	+0.03 (+0.3)	0.12 (+2.6)	0.04	0.63	10.8
APEX	45	+1.08 (+2.8)	0.17 (+1.2)	0.03	0.92	11.9

Note: ordinary least squares $R_t^s = \alpha + \beta R_t^{proxy} + \varepsilon_t$ on monthly returns, each series over its overlap with the proxy within 2000-01 to 2025-12 (n column; all windows end in 2025-12). α is in per cent per month; α and β carry Student t statistics in parentheses. σ_f/σ_p is the ratio of annualised volatilities, and TE the tracking error: the annualised standard deviation of the active return $R_t^s - R_t^{proxy}$, in per cent per year. Sources and conventions as in Table 9.

B.4 Annual properties against the other asset classes

The annual moments and scatter plots use the current U.S.-resident panel and fixed-notional hedge. The proxy is net of a 0.85% management fee and a 0.61% annual turnover deduction. The exhibits describe the proxy's relation to the other asset classes, including its behavior in equity-loss years.

Table 12: The managed-futures proxy against the other asset classes

Asset	Mean	SD	Sharpe_0	Skew	5th pctl	Max DD
Domestic equity	8.81%	18.69%	0.47	-0.24	-21.4%	-51.9%
World equity	7.42%	17.48%	0.42	-0.25	-23.1%	-48.1%
World bonds (covered)	2.82%	7.03%	0.40	+0.15	-7.6%	-35.2%
Gold	3.70%	19.70%	0.19	+1.88	-21.2%	-79.0%
Managed-futures proxy (net)	8.10%	14.18%	0.57	+0.52	-11.4%	-29.7%

Year	World equity	MF proxy	Gold	U.S. inflation
1930	-18.7%	12.7%	6.0%	-2.3%
1931	-29.5%	33.8%	10.2%	-9.0%
1937	-25.4%	-9.6%	-3.4%	3.6%
1946	-18.9%	36.3%	-15.7%	8.3%
1973	-23.5%	47.7%	58.1%	6.3%
1974	-32.2%	22.6%	47.6%	11.0%
1990	-22.5%	3.5%	-9.2%	4.9%
2002	-21.8%	4.4%	22.0%	1.6%
2008	-40.7%	25.9%	3.9%	3.8%
2022	-23.1%	9.2%	-6.0%	8.0%

Note: U.S. resident, annual real returns, 1927–2025. *Top panel:* $\text{Sharpe}_{r=0}$ is the mean over the standard deviation with a zero real cash rate, and "Max DD" is the largest peak-to-trough decline in cumulative real value. *Bottom panel:* the ten worst years for world equity and the proxy's realized return in each, where "U.S. inflation" is the same year's CPI change.

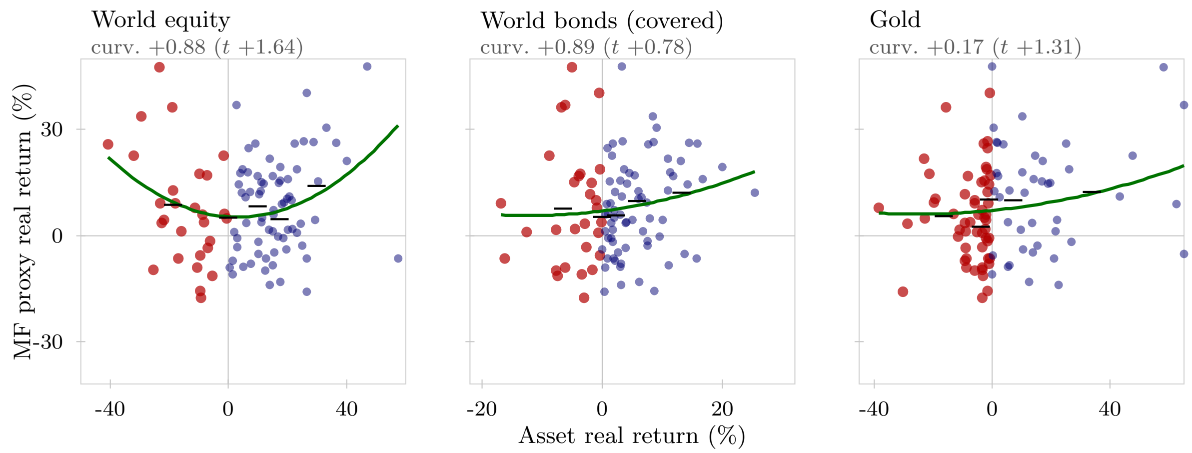

Figure 4: Annual real return of the net managed-futures proxy against the annual real return of each asset class, U.S. resident, 1927–2025. Each point is one year, with red points the years the asset itself lost and blue points the years it gained. The green curve is a descriptive quadratic fit and the black ticks are the mean proxy return within each quintile of the asset's return. "curv." is the coefficient on the squared term, and t is its heteroskedasticity-robust (White HC1) statistic.

B.5 Sensitivity to the construction choices

The series-level comparison varies signal horizons and management fees, using each variant's measured turnover cost. It describes the underlying proxy series independently of the lifecycle bootstrap and resident-currency hedge.

Table 13: Managed-futures proxy under alternative construction choices

Variant	Mean	SD	Sharpe_0	Skew	5th pctl	Corr. baseline
1/6/12 blend (baseline)†	7.91%	13.98%	0.57	+0.45	-11.8%	1.000
1/3/12 blend	7.94%	14.21%	0.56	+0.66	-11.2%	0.957
12-month signal only	8.19%	14.58%	0.56	+0.21	-14.3%	0.831
1-month signal only	5.57%	12.53%	0.44	+0.61	-10.6%	0.759
1/6/12 blend, fee doubled to 1.70%	6.98%	13.86%	0.50	+0.45	-12.6%	1.000

Note: U.S. resident, annual real returns, 1927–2025, net of each variant's management fee and its own realized turnover cost. "Corr. baseline" is the Pearson correlation of the annual series with the 1/6/12 blend used throughout the paper. The baseline mean here differs from Table 12 by about 0.19 point because that table applies a flat annual turnover drag while this one compounds the monthly realized cost. The correlations use the monthly-cost convention for every variant. †Baseline.

C Leverage ladders

The following tables use the 1,561-country-year panel, the ACO-style block continuation, and the fixed-notional forward convention. Each family has five specified gross-exposure levels. Volatility is measured after the weights are set; the tables do not estimate a volatility target. They use 10,000 paths at seed 20260827 and the same paired ACO benchmark. The benchmark has 25.28% annual volatility, 6.98% retirement ruin, and a 10% reference saving rate. Source-influence diagnostics in Table 4 examine the observations that contribute most to pooled variance.

Table 14: Proportional leverage ladder

Gross	Equity/Bonds/Gold/MF	Volatility	Saving	Ruin
100%	40/26.67/16.67/16.67	15.66%	12.35%	5.04%
125%	50/33.33/20.83/20.83	19.23%	9.46%	3.32%
150%	60/40/25/25	22.88%	7.39%	2.53%
175%	70/46.67/29.17/29.17	26.58%	5.87%	2.13%
200%	80/53.33/33.33/33.33	30.31%	4.72%	1.80%

Note: Equity is the ACO 33/67 sleeve. Bonds and managed futures use the fixed-notional hedge. Saving is the contribution rate that matches benchmark utility. Tabulated from the leverage ladder simulation dataset.

Table 15: Equal-weight leverage ladder

Gross	Equity/Bonds/Gold/MF	Volatility	Saving	Ruin
100%	25/25/25/25	15.09%	13.26%	5.90%
125%	31.25/31.25/31.25/31.25	18.50%	10.25%	4.00%
150%	37.5/37.5/37.5/37.5	21.99%	8.11%	2.97%
175%	43.75/43.75/43.75/43.75	25.54%	6.53%	2.48%
200%	50/50/50/50	29.12%	5.31%	2.27%

Note: Equity is the ACO 33/67 sleeve. Bonds and managed futures use the fixed-notional hedge. Saving is the contribution rate that matches benchmark utility. Tabulated from the leverage ladder simulation dataset.

D Historical-sample uncertainty

Monte Carlo intervals measure sampling noise conditional on the observed history and the specified model. They do not measure uncertainty about the historical return-generating process. We retain 1927–2025 as the main sample. The 1950–2025 and 1970–2025 experiments, each with 10,000 paths, address sensitivity to the historical window. They are reported in the main text and archived in the sample window directory.

The source-country omissions remove the country from resident rows, international-equity constituents, and world-bond constituents. At 5,000 paths per omission, ACO ruin ranges from 7.10% to 8.66%; the proportional 200% family ranges from 0.90% to 2.88%, and the equal-weight 200% family from 1.36% to 3.16%. The corresponding saving rates range from 4.65% to 4.89% and from 5.07% to 5.56%, against each sample's 10% reference. These are ranges across omissions, not confidence intervals, documented in the country sensitivity audit.

The separate event experiments remove Italy 1942, then Italy 1942 and France 1946. Their inputs and outputs are archived in the source exclusion directory. They measure the influence of observations identified during the audit. The exclusions apply to resident rows and global equity and bond constituents; the monthly managed-futures inputs are held fixed. The main text reports these results.

D.1 Resampling the calendar history

The outer experiment draws stationary blocks of complete calendar-year cross-sections, with mean lengths of 5, 10, or 20 years and circular wrapping at the sample endpoints. Each selected cross-section retains all available resident rows and their already constructed asset returns. Sampled years are relabelled consecutively before the inner country-block bootstrap, avoiding duplicate country-year identifiers. There are 100 outer histories for each length, each containing 99 calendar positions, with 1,000 paired inner lifecycle paths per history and a ten-year inner mean block. The outer seed is 20260905 and the inner seed is that value plus the replicate index.

The exercise preserves common shocks within each sampled calendar year and consecutive observations within outer blocks. It resamples the assembled return panel; it does not reconstruct MF signals or constituent weights from synthetic price paths. It therefore measures sensitivity to the frequency and ordering of observed states, conditional on the sleeve reconstruction. Dependence is broken at outer block joins, and country gaps still trigger the inner continuation convention. Percentile ranges include residual inner Monte Carlo noise; with 100 outer draws they are coarse historical-sensitivity estimates, not calibrated sampling-confidence intervals or probabilities that one strategy will outperform in the future. The outer-bootstrap script and replicate outputs record both specifications and all replicate-level estimates.

Table 16: Variation across resampled calendar histories at 175% exposure

Block	Family	Δruin (pp)	Saving (%)	Lower ruin (%)
5	Proportional	-5.50 [-17.51, -0.50]	5.69 [3.46, 9.22]	98
5	Equal weight	-5.05 [-18.25, 0.82]	6.25 [3.14, 12.90]	86
10	Proportional	-5.15 [-12.90, -1.19]	5.67 [3.90, 8.40]	99
10	Equal weight	-4.70 [-13.11, 0.11]	6.28 [3.51, 11.33]	94
20	Proportional	-5.05 [-11.01, -1.49]	5.91 [4.06, 8.73]	100
20	Equal weight	-4.10 [-11.04, -0.60]	6.72 [3.81, 11.74]	97

Median [5th, 95th percentile] across 100 outer histories per block length; 1,000 paired inner lifecycle paths per history. Negative differences favor diversification. Lower ruin is the fraction of simulated histories with a negative estimated difference, not a posterior probability. Ranges include residual inner Monte Carlo noise and are not calibrated confidence intervals.

E Risk-aversion sensitivity

The exercise holds portfolio weights and exposure fixed and recomputes the unlevered ACO utility target at every risk-aversion coefficient. We report both the earlier joint calibration, which varies the annual bequest coefficient with γ, and a control holding that coefficient at 2360. The latter isolates the effect of changing curvature without changing the annual bequest coefficient. Neither exercise reoptimizes portfolio weights. Ruin is a wealth-depletion event and is unchanged by preferences alone.

Table 17: Risk aversion and the saving requirement at 200% exposure

Calibration	γ	ACO	Proportional	Equal-weight
Joint calibration	2.0	10.35%	2.04%	2.58%
Joint calibration	3.0	20.09%	3.38%	3.97%
Joint calibration	3.84	15.66%	4.72%	5.31%
Joint calibration	5.0	20.25%	4.54%	5.14%
Joint calibration	7.5	20.26%	4.30%	4.91%
Joint calibration	10.0	17.60%	4.21%	4.84%
Fixed annual bequest	2.0	9.23%	2.60%	3.17%
Fixed annual bequest	3.0	12.42%	4.60%	5.18%
Fixed annual bequest	3.84	15.66%	4.72%	5.31%
Fixed annual bequest	5.0	20.26%	4.54%	5.14%
Fixed annual bequest	7.5	20.26%	4.30%	4.91%
Fixed annual bequest	10.0	17.60%	4.21%	4.84%

10,000 paired paths; ACO at 100% requires 10% saving at each γ. The joint calibration uses $\theta = 2360\,12^{3.84-\gamma}$; the fixed-annual calibration keeps $\theta = 2360$. At $\gamma = 3.84$ both coincide. All portfolios keep their weights and exposure fixed. Ruin is identical across preference calibrations; its baseline levels appear in the leverage ladders.

The modest saving response for diversified portfolios survives the fixed-coefficient control. At $\gamma = 10$, consumption accounts for virtually all of their negative expected utility: the shifted bequest term contributes less than 10^{-10} of its magnitude. For the equal-weight rule, the worst one percent of paths accounts for 92.6% of negative utility, compared with 5.3% at $\gamma = 2$. High curvature therefore does give substantial weight to adverse consumption outcomes. Social Security and the consumption floor, the distinction between upside return variation and consumption shortfalls, and the changing benchmark utility explain why this need not translate into an exploding saving requirement. The relative utility residual at the reported saving solutions is below 10^{-6} throughout the fixed-coefficient grid. Tail concentration at high γ also makes those estimates more sensitive to the sampled adverse paths.

F Contribution- and withdrawal-rate sensitivity

The policy exercise varies contribution rates at a fixed 4% withdrawal rule and withdrawal rates at a fixed 10% benchmark contribution. On the contribution axis, the comparison is anchored to benchmark utility at that row's contribution rate. On the withdrawal axis, it is anchored to the benchmark under that row's withdrawal rule. Thus saving rates should be compared within a row.

With proportional contributions and an initial-wealth-based withdrawal rule, changing the contribution rate scales account wealth and withdrawals together. This scaling leaves ruin unchanged while increasing consumption and bequests. Higher withdrawal rates increase depletion in all three portfolios. Both diversified families at 200% exposure retain lower simulated ruin and equivalent saving over this grid.

Table 18: Contribution- and withdrawal-rate sensitivity of the fixed diversified families

Rate	ACO 33/67		Proportional 200%		Equal-weight 200%	
	Ruin	Saving	Ruin	Equiv. saving	Ruin	Equiv. saving
lower is better, the ACO columns are the reference						
Contribution rate r_c (withdrawal held at 4%)						
$r_c = 5\%$	*6.98%*	*5.00%*	1.80%	2.47%	2.27%	2.74%
$r_c = 10\%^\dagger$	*6.98%*	*10.00%*	1.80%	4.72%	2.27%	5.31%
$r_c = 15\%$	*6.98%*	*15.00%*	1.80%	6.84%	2.27%	7.76%
Withdrawal rate r_w (contribution held at 10%)						
$r_w = 3\%$	*2.54%*	*10.00%*	0.92%	5.09%	1.03%	5.68%
$r_w = 4\%^\dagger$	*6.98%*	*10.00%*	1.80%	4.72%	2.27%	5.31%
$r_w = 5\%$	*15.47%*	*10.00%*	4.37%	4.00%	5.54%	4.55%

Note: 10,000 paired lifecycle paths per row, full 1,557-country-year panel, ten-year mean stationary blocks, $\phi = 30$ bp and $\kappa = 10$ bp. Entries are levels, in percent: retirement ruin and the utility-equivalent saving rate, both lower is better. The two ACO 33/67 columns are the reference. The two 200% families keep their fixed weights and gross exposure in every row. On the contribution-rate axis the ACO 33/67 benchmark also contributes r_c, so its reference saving rate is r_c itself and the family saving is the rate that matches ACO's lifecycle utility at that r_c. On the withdrawal-rate axis the benchmark saving rate stays at 10%. †Baseline.

F.1 A common planned withdrawal anchored to ACO

For each paired path i, we fix the annual real financial withdrawal at $D_i = 0.04 W_{i,65}^{\mathrm{ACO}}$, where the benchmark is unlevered ACO with a 10% saving rate. Every candidate also saves 10% and faces the same income and mortality draws. Its implied initial withdrawal rate is

$$q_{i,j} = \frac{D_i}{W_{i,65}^{j}} = 0.04 \frac{W_{i,65}^{\mathrm{ACO}}}{W_{i,65}^{j}}.$$

Thus a candidate that accumulates less capital than ACO on a particular path must withdraw more than 4% of its own capital to finance the target. For example, capital equal to two-thirds of ACO's implies a 6% initial rate. Lower annual volatility alone does not determine this capital ratio. Among the 9,832 paths reaching retirement with a positive benchmark target, the proportional and equal-weight 175% rules have median implied rates of 2.69% and 2.93%; their 90th percentiles are 4.72% and 6.83%. Rates exceed 4% on 18.46% and 32.26% of these paths, respectively, and exceed 6% on 2.72% and 14.14%. Failure to fund the common financial draw occurs on 7.10% of these paths for unlevered ACO, 2.21% for proportional 175%, and 8.12% for equal-weight 175%. These probabilities condition on retirement and a positive target, unlike the main unconditional ruin statistics. On the common denominator of all 10,000 simulated lifetimes, the corresponding frequencies are 6.98%, 2.17%, and 7.98%. ACO has exactly the same 698 failures as in the main experiment: 698/10,000 = 6.98% and 698/9,832 = 7.10% answer different conditioning questions. Replacing the common target with each strategy's own 4%-of-capital target reproduces its main-engine failure indicators path by path for all four tested strategies; the recorded audit also verifies equality with their archived main ruin rates. The corrected funding ratios exclude paths with no retirement years. This does not revise the main experiment's definition of unconditional ruin.

The target remains fixed if ACO subsequently depletes its account; public pensions continue under the common household assumptions. The equal-weight result therefore identifies a limitation in financing ACO's planned income across heterogeneous accumulation outcomes, rather than its failure rate under a 4%-of-own-capital policy. The reported rate distribution describes all eligible paths and does not by itself attribute the difference in failures to high initial withdrawal rates. The common-target script and numerical outputs record the experiment on 10,000 paired paths at seed 20260827.

Table 19: Common ACO planned income and implied withdrawal rates

Portfolio	Fail, all	Fail, retired	Median draw	P90 draw	Mean deficit
ACO	6.98	7.10	4.00	4.00	1.98
ACO 175%	6.25	6.36	2.22	4.67	2.40
Proportional 175%	2.17	2.21	2.69	4.72	0.73
Equal 175%	7.98	8.12	2.93	6.83	3.23

All entries in percent. All-path failures use 10,000 lifetimes; retired failures, initial draws and mean deficit use 9,832 eligible paths. Draw is the target divided by own initial retirement capital. Mean deficit is the mean path-level share of cumulative planned financial withdrawals left unfunded, including zero for successful paths. Public pensions are common across strategies and excluded from this financial-deficit measure.

G Reproducibility and Conversion Checks

The baseline is identified by 1,561 resident country-years, 10,000 paths, seed 20260827, ten-year mean blocks, and the fixed-notional hedge convention. Portfolios share lifetime draws within each historical panel; changing the panel changes the draws even at the same seed. Full-panel numerical outputs are archived in the core results and leverage ladder datasets. The constant-exposure sleeve ablations are stored in the ablation dataset; the replication script records the reallocation rule and all simulation settings. Historical-window outputs and replication commands are recorded in the sample window directory. The variance concentration script checks baseline volatility agreement before reporting concentration and deletion diagnostics.

The annual conversion and simulation can be reproduced from the supplied annual inputs. The monthly managed-futures reconstruction has an upstream engine dependency outside this paper directory. The package consumes the frozen monthly output of that engine. Reconstructing it from provider data requires the upstream engine, while regenerating the external-index comparisons requires the licensed inputs.

For a foreign asset's local real return R_f^L and inflation π_f, the nominal gross return is $(1 + R_f^L)(1+\pi_f)$. A fixed-notional forward covers the initial foreign principal. If q is the realized resident-currency value of one unit of foreign currency relative to its initial value, G_f the foreign nominal gross asset return, and B_c/B_f the covered-interest-parity forward factor, the hedged nominal gross payoff is

$$G^H_{f\to c} = \frac{B_c}{B_f} + (G_f - 1)q.$$

Deflating by resident inflation produces the real return used in the panel. The second term retains currency exposure on the foreign gain or loss. Consequently, equality of all residents' asset excess returns is not a valid check for this convention. That identity belongs to the earlier idealized full-terminal-value hedge.

The return-cap diagnostic, ideal hedge, legacy block restart, restricted historical windows, and source exclusions are distinct experiments. Their output files identify the changed convention. The baseline retains large returns. Documented market-year exclusions and source-influence diagnostics are recorded separately.

H Margin calls on the do-it-yourself route

Direct portfolio borrowing introduces a maintenance-margin constraint. We evaluate six directly financed allocations using the 1,561-country-year panel, fixed-notional currency hedges, and the baseline fees and costs. The account rebalances to its target exposure at the start of each year, and its equity share is tested against a 25% maintenance threshold at year-end, before the next rebalance. We draw 10,000 paired 86-year return paths with ten-year mean blocks, ACO-style continuation, and seed 20260827. This account diagnostic follows each return path for the full horizon, without household contributions, withdrawals, or mortality truncation.

For initial equity normalized to one, gross assets L, debt $L-1$, asset-book nominal gross return G_A, and debt gross accrual G_D, the year-end equity fraction is

$$m_{\text{end}} = \frac{LG_A - (L-1)G_D}{LG_A}.$$

A maintenance call at threshold m occurs when $G_A < (L-1)G_D/[L(1-m)]$, provided the asset value remains positive. The threshold therefore depends on financing as well as exposure. Nominal rather than real account values determine the margin test. To match the paper's real financing deduction exactly, debt accrues at $G_D = (1 + R_f^{(c)} + \phi)(1 + \pi_c)$. The asset book includes sleeve fees and hedge costs, while the borrowing charge enters the debt leg.

Table 20: Year-end maintenance-margin diagnostic

Allocation	Gross exposure	Threshold breaches	Minimum equity share
ACO 33/67	175%	1,081	21.41%
ACO 33/67	200%	17,934	8.31%
Equal-weight four-sleeve	175%	0	48.91%
Proportional four-sleeve	200%	0	38.59%
Equal-weight four-sleeve	200%	0	40.39%
90/60 covered global bonds	150%	0	53.24%

Note: 860,000 simulated account-years per allocation; maintenance threshold 25%. Minimum equity share is the smallest year-end ratio of account equity to gross assets. Account-years reuse the historical panel through block resampling. This table can be reproduced from the margin simulation dataset using the margin call experiment script.

The four diversified allocations have no year-end threshold breaches. ACO records 1,081 at 175% exposure and 17,934 at 200%, or 0.13% and 2.09% of account-years. Exposure is reset annually in this diagnostic, including after a breach; it measures threshold crossings rather than a lifecycle policy with forced liquidation. Annual rebalancing restores the initial equity cushion: 50% of assets at 200% gross exposure and 57.14% at 175%.

H.1 Monthly monitoring between annual rebalances

We supplement the annual resident-panel experiment with a monthly USD proxy control over 1970–2025. Equity uses 33% US and 67% ex-US total-return simulations. A US Treasury total-return simulation substitutes for the global bond sleeve, while gold and the net MF series provide the other two sleeves. This control has 56 complete calendar years. Its market composition and numeraire are explicit so that frequency effects can be measured within the same monthly dataset. It is not a monthly reconstruction of the 16-resident annual panel.

For sleeve weights w_j, exposure L, and month m within a year,

$$A_m = L \sum_j w_j \prod_{s=1}^{m} (1 + r_{j,s}), \qquad D_m = (L-1) \prod_{s=1}^{m} (1 + r_{f,s} + 0.003/12), \qquad e_m = 1 - D_m/A_m.$$

Positions are held through the year; no monthly reset restores the equity cushion. We compare $\min_m e_m$ with the December value on these same paths, at maintenance thresholds of 25%, 35%, and 40%. The replication also resamples 10,000 paired 86-year paths in blocks averaging ten years. Crossings are recorded without imposing liquidation, allowing the monthly and annual monitoring rules to be compared without changing the return path after a call.

At 200% exposure and a 40% maintenance requirement, ACO crosses in 13 historical years under monthly monitoring, against nine at year-end. The proportional and equal-weight portfolios stay above that threshold in this sample, with minimum month-end equity shares of 40.3% and 41.2%. At the baseline 25% requirement, ACO crosses in 2008 under both monitoring frequencies and the diversified rules have no crossing. Monthly observation thus detects additional funding pressure when the maintenance requirement is tighter. Intramonth extrema and changes in broker requirements require finer data; neither the annual nor monthly experiment measures them.

I Full comparisons under alternative panel treatments

The main text reports all portfolio rules under both the raw and source-screened panels. The following tables give the corresponding complete return moments and the separate 0.5% winsorization sensitivity. These treatments were chosen after the influence audit. Excluded observations remain available in the raw panel; the screen is not treated as proof that excluded economic losses did not occur. The exclusion list, reasons, quantile thresholds, and input hashes are recorded in the panel-comparison manifest.

Table 21: Month-end and year-end maintenance tests on identical USD proxy paths

Portfolio	Gross (%)	Maintenance	Monthly years	Annual years	Min. equity share
ACO	100	25.00%	0	0	100.00%
ACO	100	40.00%	0	0	100.00%
ACO	125	25.00%	0	0	62.51%
ACO	125	40.00%	0	0	62.51%
ACO	150	25.00%	0	0	37.52%
ACO	150	40.00%	1	0	37.52%
ACO	175	25.00%	1	1	19.67%
ACO	175	40.00%	3	2	19.67%
ACO	200	25.00%	1	1	6.29%
ACO	200	40.00%	13	9	6.29%
Proportional	100	25.00%	0	0	100.00%
Proportional	100	40.00%	0	0	100.00%
Proportional	125	25.00%	0	0	76.14%
Proportional	125	40.00%	0	0	76.14%
Proportional	150	25.00%	0	0	60.23%
Proportional	150	40.00%	0	0	60.23%
Proportional	175	25.00%	0	0	48.86%
Proportional	175	40.00%	0	0	48.86%
Proportional	200	25.00%	0	0	40.34%
Proportional	200	40.00%	0	0	40.34%
Equal-weight	100	25.00%	0	0	100.00%
Equal-weight	100	40.00%	0	0	100.00%
Equal-weight	125	25.00%	0	0	76.47%
Equal-weight	125	40.00%	0	0	76.47%
Equal-weight	150	25.00%	0	0	60.79%
Equal-weight	150	40.00%	0	0	60.79%
Equal-weight	175	25.00%	0	0	49.58%
Equal-weight	175	40.00%	0	0	49.58%
Equal-weight	200	25.00%	0	0	41.18%
Equal-weight	200	40.00%	0	0	41.18%

1970–2025, 56 complete years. Entries count historical years with at least one crossing. Sleeve quantities are fixed within each year; exposure resets annually. VTI/VXUS total-return simulations supply 33/67 equities; IEF supplies a US Treasury proxy for global bonds; gold is spot less custody, MF is the reconstructed net USD series. Debt accrues USD cash plus 30 bp/year. The two tests use the same asset and debt paths. Monthly closes do not measure intramonth extrema or simulate forced liquidation. Bootstrap outputs and the 35% threshold are provided in the replication dataset.

Table 22: Complete annual return moments under both panels

Portfolio	Raw mean	Raw vol.	Raw 1st pct.	Screen mean	Screen vol.	Screen 1st pct.
Domestic equity	7.60%	23.34%	-45.64%	8.04%	22.70%	-41.22%
ACO 33/67	8.12%	25.28%	-40.36%	7.91%	17.02%	-38.12%
Stocks/I	7.99%	22.59%	-40.22%	7.95%	17.61%	-38.18%
Domestic balanced	5.38%	15.82%	-34.19%	5.83%	15.02%	-27.11%
Balanced/I	5.61%	15.04%	-27.63%	5.77%	12.06%	-23.74%
90/60 local	8.25%	23.95%	-36.55%	8.15%	16.97%	-32.81%
90/60 global bonds	8.19%	24.13%	-36.19%	8.06%	16.47%	-32.03%
ACO, constant real FX	7.44%	15.75%	-37.40%	7.79%	15.66%	-37.47%
ACO 33/67 125%	10.00%	31.47%	-48.19%	9.65%	21.15%	-47.71%
ACO 33/67 150%	11.89%	37.71%	-57.47%	11.39%	25.33%	-56.37%
ACO 33/67 175%	13.78%	43.99%	-66.32%	13.14%	29.53%	-65.98%
ACO 33/67 200%	15.67%	50.28%	-75.74%	14.88%	33.74%	-75.46%
60/40 ACO/covered bonds	5.64%	16.51%	-25.67%	5.69%	11.34%	-22.62%
60/40 ACO + 33.33 MF	7.91%	17.95%	-20.08%	7.89%	12.69%	-17.75%
60/40 ACO + 33.33 gold	6.79%	23.86%	-23.33%	6.47%	13.18%	-20.36%
54/36/20/20 ACO	7.18%	19.94%	-19.08%	7.00%	11.92%	-16.28%
Proportional 100%	5.66%	15.66%	-16.09%	5.60%	9.43%	-13.33%
Proportional 125%	6.93%	19.23%	-18.63%	6.76%	11.44%	-15.42%
Proportional 150%	8.20%	22.88%	-21.18%	7.92%	13.53%	-18.34%
Proportional 175%	9.48%	26.58%	-24.40%	9.08%	15.66%	-21.67%
Proportional 200%	10.75%	30.31%	-27.72%	10.25%	17.81%	-24.43%
Equal-weight 100%	5.35%	15.09%	-15.53%	5.27%	9.19%	-12.80%
Equal-weight 125%	6.55%	18.50%	-15.83%	6.35%	11.12%	-14.27%
Equal-weight 150%	7.75%	21.99%	-18.86%	7.44%	13.12%	-15.90%
Equal-weight 175%	8.95%	25.54%	-21.16%	8.52%	15.17%	-18.97%
Equal-weight 200%	10.15%	29.12%	-24.18%	9.60%	17.25%	-21.93%
90/60/25/25 ACO	10.75%	30.34%	-29.39%	10.29%	18.27%	-26.45%

10,000 paths per panel, seed 20260827, 1927–2025. Screened sources exclude documented disruptions, inflation outside $(-20\%, 50\%)$, and Italy 1942 identified by the audit. The same exclusions apply to resident rows and reconstructed equity and bond baskets. The screen is retrospective and does not establish a historically implementable trading rule. MF inputs are unchanged. Saving matches each panel's own unlevered ACO at 10%; cross-panel saving levels therefore have different utility targets. Common seeds pair strategies within each panel, not identical paths across different panel supports.

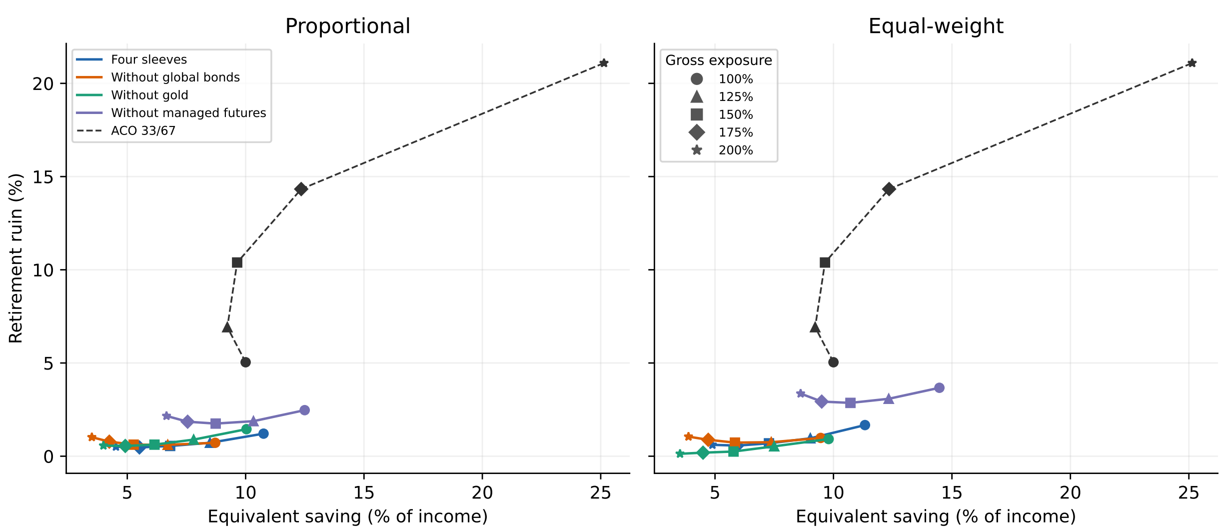

Figure 5: Complete and ablated exposure ladders on the source-screened panel. Marker shapes identify 100, 125, 150, 175, and 200% exposure. All compositions use 10,000 shared paths and the screened unlevered ACO utility target. The raw-panel counterpart appears in Figure 1.

Table 23: Portfolio comparison after 0.5% sleeve winsorization

Portfolio	Volatility	Ruin	Saving
Domestic equity	22.76%	14.74%	16.74%
ACO 33/67	17.03%	7.35%	10.00%
Stocks/I	17.61%	7.90%	10.47%
Domestic balanced	15.48%	13.33%	18.64%
Balanced/I	12.52%	9.81%	14.92%
90/60 local	17.13%	7.85%	9.79%
90/60 global bonds	16.48%	6.56%	9.11%
ACO, constant real FX	15.65%	4.33%	9.03%
ACO 33/67 125%	21.00%	8.18%	8.40%
ACO 33/67 150%	25.05%	10.33%	7.94%
ACO 33/67 175%	29.15%	14.07%	8.87%
ACO 33/67 200%	33.27%	19.43%	13.37%
60/40 ACO/covered bonds	11.63%	8.62%	13.90%
60/40 ACO + 33.33 MF	12.81%	2.34%	7.32%
60/40 ACO + 33.33 gold	13.40%	4.48%	10.58%
54/36/20/20 ACO	12.14%	2.73%	8.73%
Proportional 100%	9.83%	4.87%	12.06%
Proportional 125%	11.70%	3.01%	9.15%
Proportional 150%	13.70%	1.91%	7.09%
Proportional 175%	15.79%	1.35%	5.60%
Proportional 200%	17.92%	0.99%	4.49%
Equal-weight 100%	9.65%	5.57%	12.83%
Equal-weight 125%	11.47%	3.62%	9.85%
Equal-weight 150%	13.43%	2.38%	7.73%
Equal-weight 175%	15.47%	1.58%	6.17%
Equal-weight 200%	17.57%	1.23%	5.01%
90/60/25/25 ACO	18.30%	1.29%	4.66%

Each of the five risky sleeve return columns is clipped at its pooled 0.5th and 99.5th percentiles before leverage and costs. Cash and inflation are unchanged. All 1,561 resident observations remain. Quantiles use the full sample; this is a statistical sensitivity, not an investability screen. Each portfolio is evaluated against the winsorized ACO reference.

J Matched-leverage comparisons

ACO retains its 33/67 domestic–international equity mix at each exposure. All borrowed amounts use the resident bill and the same financing spread as the diversified rules. These comparisons hold borrowing constant; portfolio volatility remains an outcome of asset composition.

Table 24: Fixed leverage ladder for ACO 33/67

Gross	Volatility	Equiv. saving	Ruin
100%	25.28%	10.00%	6.98%
125%	31.47%	8.56%	8.13%
150%	37.71%	8.30%	10.69%
175%	43.99%	9.64%	14.81%
200%	50.28%	15.66%	20.28%

10,000 paired paths, 1927–2025. Borrowing costs the resident bill plus the same 30 bp real deduction used for the diversified portfolios. The utility target remains ACO at 100% exposure and 10% saving.

Table 25: Levered ACO in the household-policy sensitivities

Saving	Withdrawal	ACO ruin	ACO equiv.	Prop. equiv.	Equal equiv.
5.00%	4.00%	20.28%	4.63%	2.47%	2.74%
10.00%	4.00%	20.28%	15.66%	4.72%	5.31%
15.00%	4.00%	20.28%	46.06%	6.84%	7.76%
10.00%	3.00%	13.73%	9.84%	5.09%	5.68%
10.00%	4.00%	20.28%	15.66%	4.72%	5.31%
10.00%	5.00%	26.75%	24.11%	4.00%	4.55%

All three portfolios have 200% exposure. The reference is unlevered ACO at the contribution and withdrawal rates in each row. The diversified ruin rates are reported in Table 18.

Table 26: Financing and hedge-cost sensitivities at matched exposure

Spread	Hedge cost	ACO ruin	ACO equiv.	Prop. ruin	Equal ruin
0.30%	0.10%	20.28%	15.66%	1.80%	2.27%
1.00%	0.10%	22.73%	22.03%	2.44%	3.16%
2.00%	0.10%	26.38%	38.35%	3.82%	4.91%
3.00%	0.10%	29.73%	76.63%	6.19%	8.13%
0.30%	0.10%	20.28%	15.66%	1.80%	2.27%
0.30%	0.50%	20.28%	15.66%	2.08%	2.67%
0.30%	1.00%	20.28%	15.66%	2.57%	3.51%

All three portfolios have 200% exposure. ACO equities are unhedged, so hedge-cost changes affect only the diversified sleeves. Equivalent saving targets unlevered ACO at 10% saving.

Table 27: Construction checks including levered ACO

Specification	ACO 100%	ACO 200%	Prop. 200%	Equal 200%
Baseline	6.98%	20.28%	1.80%	2.27%
Block restart	7.38%	19.68%	1.84%	2.26%
Full-value hedge	6.98%	20.28%	1.82%	2.27%
Return screen	7.38%	21.71%	1.57%	2.02%
War screen	6.04%	25.67%	0.30%	0.27%
Source-country range	7.10–8.66%	19.92–24.00%	0.90–2.88%	1.36–3.16%

Retirement ruin; 10,000 paired paths for construction checks and 5,000 per source-country omission. All other assumptions follow the corresponding diagnostic.

K Post-1970 sleeve roles

This window uses the current raw panel restricted to 1970–2025, with the same fixed-notional hedge, financing, fees and bootstrap continuation as the main experiment. Both complete families and each pro-rata ablation hold 175% gross exposure. It complements the full-history ablations by evaluating the compositions after the administered-gold era. Differences across windows also reflect the bond and equity regimes and changes in the MF market universe. The proportional rule's bond ablation illustrates a trade-off: removing bonds raises volatility and estimated ruin while lowering equivalent saving. Gold's removal worsens saving and ruin for both families, although the equal-weight saving difference is small. These findings support distinct capital-preservation roles without establishing that every sleeve improves every metric. The ablation script, with `--year-from 1970`, produces the recorded window results.

Table 28: Complete and ablated 175% portfolios, 1970–2025

Portfolio	Volatility	Saving	Ruin
ACO 33/67	17.24	10.00	6.67
ACO 33/67 175%	29.79	12.51	16.16
Proportional	15.03	4.16	0.04
Proportional without bonds	19.30	2.87	0.20
Proportional without gold	16.03	4.63	0.64
Proportional without MF	17.14	5.48	0.68
Equal weight	14.81	4.16	0.03
Equal weight without bonds	19.09	3.01	0.16
Equal weight without gold	13.27	4.19	0.08
Equal weight without MF	16.90	6.14	0.81

All numbers in percent; 10,000 paired paths, 894 resident country-years, fixed-notional hedges and baseline costs. Removing a sleeve reallocates its weight pro rata at unchanged gross exposure. Ruin rates of 0.03% and 0.04% correspond to three and four paths; their difference alone does not establish a ranking. The ACO utility target is recomputed on this window.

L Return and financing stresses at 175% exposure

These two experiments vary one assumption at a time, retain fixed portfolio weights and common path draws, and keep the utility target at unlevered ACO saving 10%. MF deductions are additional to the 0.85% management fee and baseline measured turnover cost. A deduction d reduces each resident-year net MF return by d, and hence portfolio return by its MF weight times d. This leaves annual return variance unchanged within each fixed portfolio; it does not emulate a change in signals or in crisis dependence. It is a transparent test of tolerance to lower sleeve returns. The tested deductions bracket the equal-weight saving crossing between 300 and 400 bp and its ruin crossing between 400 and 600 bp. The proportional rule's saving crossing lies between 600 and 700 bp; its ruin remains below ACO at the largest deduction tested. These are grid brackets, not exact break-even costs or estimates of the proxy's measurement error. At 175% gross exposure, a 100 bp spread costs 75 bp of investor capital annually in addition to the resident bill financing leg. A 300 bp spread costs 225 bp. The 100 bp case already represents costly financing relative to the efficient retail routes discussed in Section 3.3; 200–300 bp are deliberately severe stresses. The proportional saving crossing lies between 250 and 300 bp; the equal-weight crossing lies between

Table 29: Annual managed-futures return deductions at 175% exposure

Deduction (bp)	Prop. saving	Prop. ruin	Equal saving	Equal ruin
0	5.87	2.13	6.53	2.48
100	6.35	2.32	7.35	3.11
200	6.89	2.57	8.30	3.88
300	7.47	3.01	9.39	4.76
400	8.10	3.39	10.67	5.92
600	9.55	4.65	13.95	9.12
700	10.39	5.36	16.07	11.24

Saving and ruin in percent. Each deduction is applied after baseline MF fees and turnover costs, on every resident-year, without rescaling fluctuations. 10,000 paired paths per specification; unlevered ACO has 10% saving and 6.98% ruin throughout. These are annual return stress tests, not proportional reductions in an estimated risk premium.

Table 30: Financing and hedge-cost stress at 175% exposure

Spread (bp)	Hedge (bp)	Prop. saving	Prop. ruin	Equal saving	Equal ruin
30	10	5.87	2.13	6.53	2.48
100	10	6.78	2.51	7.53	3.24
200	10	8.35	3.59	9.29	4.68
250	10	9.28	4.44	10.36	5.65
300	10	10.34	5.32	11.58	6.92
30	100	7.08	2.73	8.10	3.76
30	50	6.38	2.35	7.18	2.93

Saving and ruin in percent; 10,000 paired paths per row. The spread is additional to the resident bill return and applies to 75% of capital. Hedge friction applies to covered sleeve notionals. Unlevered ACO is unaffected by these charges and retains 10% saving and 6.98% ruin. The grid brackets crossings; no interpolated break-even estimate is reported.

200 and 250 bp. No ruin crossing is established on the financing grid through 300 bp, although the equal-weight estimate there is only 0.06 percentage point below ACO. A hedge friction of 100 bp separately leaves both families below the benchmark in saving and ruin. These controls do not impose a forced-liquidation policy; the monthly maintenance-margin diagnostic remains a separate threshold experiment. Replication outputs: MF deductions; financing and hedge costs.